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ABSTRACT

This project, titled “**AWS 3-Tier Architecture Design & Deployment,**” demonstrates the implementation of a **scalable, highly available, and production-ready web application** using **Amazon Web Services (AWS)**. The system is designed using a **three-tier architecture**, consisting of the **presentation layer, application layer, and data layer**, ensuring clear separation of concerns and improved system performance.

The project utilizes **React.js** for the frontend, **Node.js** for backend processing, and **MySQL on Amazon RDS** for data storage. Key AWS services such as **EC2, Application Load Balancer, Auto Scaling Groups, CloudFront, Route53, and CloudWatch** are integrated to achieve **load balancing, scalability, monitoring, and optimized content delivery**. This implementation ensures **high availability, fault tolerance, and efficient resource utilization**, while providing hands-on experience in **cloud infrastructure design and deployment**.

INTRODUCTION

With the increasing demand for modern web applications, it has become essential to build systems that are **scalable, secure, and highly available**. Traditional **monolithic architectures** often face issues such as **poor scalability, performance limitations, and lack of fault tolerance** under heavy traffic. To address these challenges, this project implements a **cloud-based three-tier architecture on AWS**, which separates the system into **presentation, application, and data layers**, improving overall system design.

The **presentation layer** manages user interaction, the **application layer** handles business logic, and the **data layer** stores information using **Amazon RDS**. By leveraging AWS services such as **Amazon EC2, Application Load Balancer, Auto Scaling, CloudFront, Route53, and CloudWatch**, the system achieves **high availability, fault tolerance, and efficient performance**, making it capable of handling dynamic workloads effectively.

2.1 MOTIVATION

The motivation behind this project stems from the need to create scalable and robust web applications capable of handling varying workloads without compromising performance or security. Traditional single-server deployments are often unable to manage high traffic efficiently, leading to slow response times, potential downtime, and limited fault tolerance. By implementing a cloud-based three-tier architecture, this project aims to demonstrate how scalable, resilient, and highly available systems can be designed while gaining hands-on experience with modern cloud technologies and best practices in enterprise application deployment.

2.2 PROBLEM STATEMENT

Traditional application deployments, relying on single-server architectures, frequently encounter limitations such as poor scalability, increased vulnerability to failures, and difficulties in maintenance. These architectures lack proper segregation of functional components, which complicates performance optimization, security management, and resource allocation. The problem addressed in this project is how to design and implement a **scalable, secure, and fault-tolerant system** using a three-tier architecture where each layer performs specialized functions while communicating efficiently to ensure optimal performance and high availability.

2.3 OBJECTIVE AND GOALS

The primary objective of this project is to design and deploy a **scalable, secure, and highly available three-tier architecture** using cloud services. The system separates functionality into web, application, and database layers to enhance performance, maintainability, and reliability.

The goals include deploying the application on **EC2 instances**, integrating **Application Load Balancers** for efficient traffic distribution, implementing **Auto Scaling** to handle varying user loads, and ensuring secure communication across all layers. Additionally, the project emphasizes the use of **CloudWatch** for monitoring and logging to provide insights into system health and operations, while gaining practical experience in cloud-based deployment and modern enterprise architecture.

2.4 DESCRIPTION OF TECHNOLOGY USED

The project is developed using modern web and cloud technologies to ensure **scalability and performance**. **React.js** is used for building a responsive

and user-friendly frontend, while **Node.js** handles backend logic and API requests efficiently. **Amazon EC2** instances are used to host both the presentation and application tiers, and **Amazon RDS (MySQL)** is used for **secure data storage and management**.

For networking and deployment, **Amazon VPC** is used to create an **isolated environment** with public and private subnets, while **Application Load Balancers (ALB)** distribute traffic across instances. **Auto Scaling Groups** ensure **high availability** by automatically adjusting resources based on demand, and **Amazon CloudWatch** is used for **monitoring and triggering scaling actions**. Additionally, **Amazon Route53** manages DNS, **AWS Certificate Manager (ACM)** enables **HTTPS**, and **Amazon CloudFront** improves performance and provides **secure global access** to the application.

2.5 SCOPE AND LIMITATION

The scope of this project encompasses the complete design and deployment of a cloud-based three-tier architecture, including web, application, and database layers. It demonstrates scalability, high availability, secure communication, and fault tolerance through the integration of AWS services. Users can observe practical applications of load balancing, auto-scaling, content delivery optimization with **CloudFront**, and centralized logging with **CloudWatch**.

However, this project is implemented as a demonstration and does not include advanced production-level features such as fully automated **CI/CD pipelines**, extensive security measures like **Web Application Firewalls (WAF)**, or real-time analytics dashboards. Additionally, cost constraints and free-tier limitations may restrict full-scale deployment, and further optimization would be necessary for enterprise-level production use.

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM & NEED FOR SYSTEM

In many traditional web application deployments, a **single-server or monolithic architecture** is used, where the user interface, business logic, and database reside on the same server or tightly coupled servers. While this approach is simple to implement, it faces significant limitations as application traffic grows. The system often struggles to handle **high traffic volumes**, leading to slow response times, downtime, and poor user experience. Additionally, these architectures are difficult to scale, maintain, and secure, and a failure in one component can impact the entire system.

There is a clear need for a **scalable, reliable, and secure architecture** that can efficiently manage increasing workloads while maintaining system performance. By separating the system into distinct layers for presentation, application logic, and data storage, it becomes easier to **scale components independently**, improve fault tolerance, and monitor performance effectively. Cloud-based solutions like **AWS** provide the tools and infrastructure to achieve these goals, enabling robust deployment for enterprise-grade applications.

3.2 SCOPE AND LIMITATION OF EXISTING SYSTEM

The existing single-server systems or traditional web architectures typically allow only **limited traffic handling capacity** and lack proper resource management. They do not support **auto-scaling or load balancing**, making them prone to failures under peak loads. Security measures are often minimal, and monitoring capabilities are limited, which increases the risk of downtime and undetected issues.

The limitations of existing systems include the inability to support **redundancy, high availability, or fault tolerance**, difficulty in maintaining and updating the application without downtime, and the challenge of efficiently distributing workload across resources.

3.3 PROJECT FEATURES

This project introduces a **modern, cloud-based three-tier architecture** designed to overcome the limitations of traditional systems. Key features include:

- **Presentation Layer:** A responsive and user-friendly frontend built with React.js for seamless user interaction.
- **Application Layer:** Node.js backend managing business logic and processing requests efficiently.
- **Database Layer:** Managed MySQL database on AWS RDS for secure and scalable data storage.
- **Auto Scaling & Load Balancing:** Automatic adjustment of computing resources using EC2 Auto Scaling Groups and Application Load Balancer to handle variable workloads.
- **Content Delivery Optimization:** CloudFront CDN for fast and reliable content distribution.
- **Monitoring and Logging:** CloudWatch integration for real-time metrics, centralized logs, and proactive issue detection.
- **Secure Architecture:** HTTPS communication using SSL certificates and secure access controls via AWS IAM.

These features ensure that the deployed system is highly available, scalable, and maintainable while providing an enterprise-grade user experience.

3.4 PROPOSED SYSTEM

The proposed system replaces the traditional monolithic architecture with a **cloud-based three-tier deployment**. It divides the system into **presentation, application, and database layers**, each running on dedicated AWS resources.

- The **presentation layer** runs on EC2 instances behind a **CloudFront CDN** to accelerate content delivery.
- The **application layer** is managed using **EC2 instances within an Auto Scaling Group**, ensuring that requests are distributed evenly through an **Application Load Balancer**.
- The **database layer** utilizes **AWS RDS (MySQL)** for reliable and secure data storage.
- All system logs and metrics are collected in **CloudWatch**, enabling centralized monitoring and easier troubleshooting.

This architecture provides **scalability, high availability, fault tolerance, and security**, allowing the system to handle varying workloads efficiently while ensuring optimal performance and maintainability.

3.5 REQUIREMENT ANALYSIS

The requirements for implementing this project are divided into **functional** and **non-functional** categories.

Functional Requirements:

- Deploy a three-tier architecture on AWS with separate layers for frontend, backend, and database.
- Implement CRUD operations for data management.
- Configure Auto Scaling and Load Balancing to handle dynamic traffic.
- Integrate CloudWatch for logging and monitoring application performance.

- Optimize content delivery using CloudFront CDN.

Non-Functional Requirements:

- Ensure high availability and fault tolerance across all layers.
- Maintain secure communication with SSL and proper IAM access controls.
- Ensure scalability to handle varying user loads efficiently.
- Provide centralized monitoring and logging for system health and troubleshooting.
- Ensure performance optimization with minimal latency and fast response times.

This requirement analysis ensures that the proposed system addresses the limitations of traditional architectures while providing a **robust, scalable, and secure cloud deployment** suitable for enterprise-level applications.

SYSTEM DESIGN

The system design for this project focuses on creating a robust, scalable, and highly available web application using a three-tier architecture deployed on Amazon Web Services (AWS). The design ensures clear separation of concerns among the presentation, application, and data layers while integrating essential cloud services for security, monitoring, and performance optimization.

4.1 TABLE SPECIFICATIONS

The project uses a relational database structure to store and manage data efficiently. The main tables designed for the application include **Author** and **Book**.

- **Author Table:** Contains information about authors, including their ID, name, date of birth, biography, and timestamps for creation and updates.
- **Book Table:** Stores details about books, including book ID, title, release date, description, number of pages, creation and update timestamps, and a foreign key referencing the associated author.

These tables are structured to maintain data integrity and support the one-to-many relationship between authors and books.

4.2 ARCHITECTURE DIAGRAM

The architecture diagram represents a **cloud-based three-tier system on AWS**, consisting of the **presentation layer, application layer, and data layer**, each handling specific functions. User requests are routed through **Amazon Route53** and delivered via **Amazon CloudFront** to improve **performance and reduce latency**, while an **Application Load Balancer (ALB)** distributes traffic

across multiple **EC2 instances** within an **Auto Scaling Group** to ensure **high availability**.

The **EC2 instances** process application logic and communicate with the **Amazon RDS (MySQL)** database for data operations. **Amazon CloudWatch** is used for **monitoring and logging** system activities. This architecture provides **scalability, fault tolerance, and reliable performance** through the integration of multiple AWS services

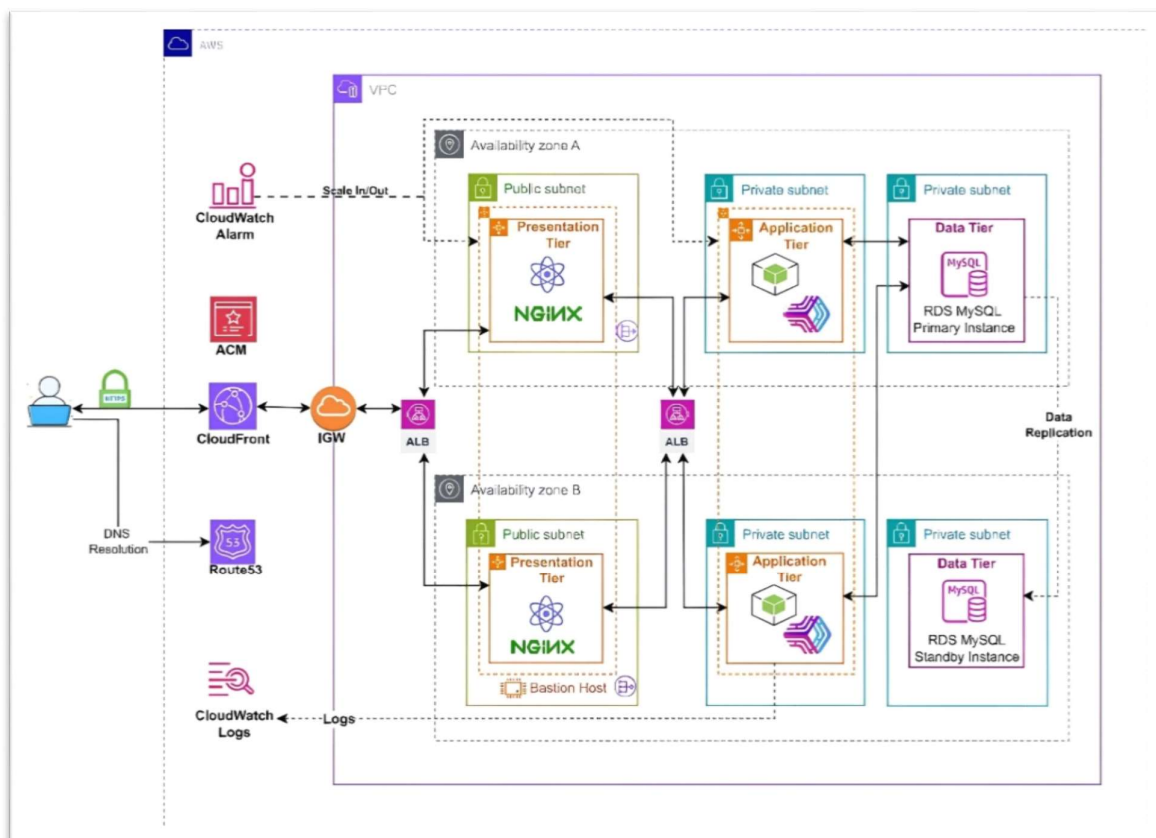


Fig: Architecture Diagram

4.3 ER DIAGRAM

The Entity Relationship Diagram represents the database structure and relationships between entities in the system.

The system includes two main entities: **Author** and **Book**. Each Author has attributes such as ID, name, date of birth, and biography, while each Book contains attributes like title, release date, description, and number of pages.

A one-to-many relationship exists between Author and Book, indicating that one author can write multiple books, while each book is associated with a single author. This relationship is implemented using a foreign key in the Book table referencing the Author table.

The ER diagram helps in visualizing the logical structure of the database and ensures efficient data organization.

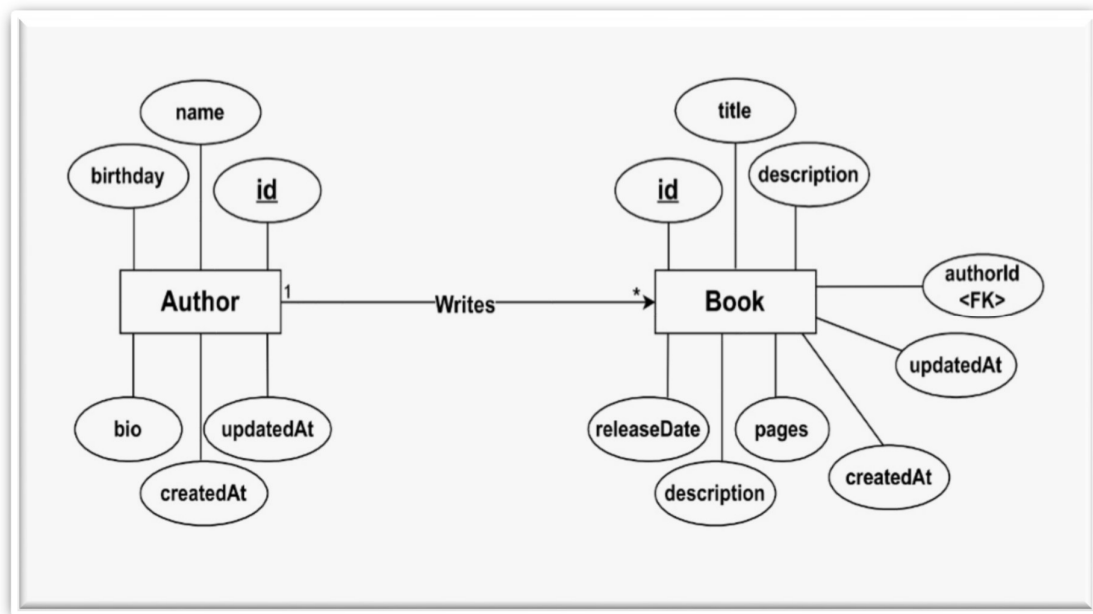


Fig: ER Diagram

4.4 SEQUENCE DIAGRAM

The sequence diagram shows how a user request flows through the system. The request is sent to CloudFront, forwarded to the Application Load Balancer, and then routed to an EC2 instance for processing.

The EC2 instance interacts with the RDS database and sends logs to CloudWatch. After processing, the response is returned back to the user through the load balancer and CloudFront.

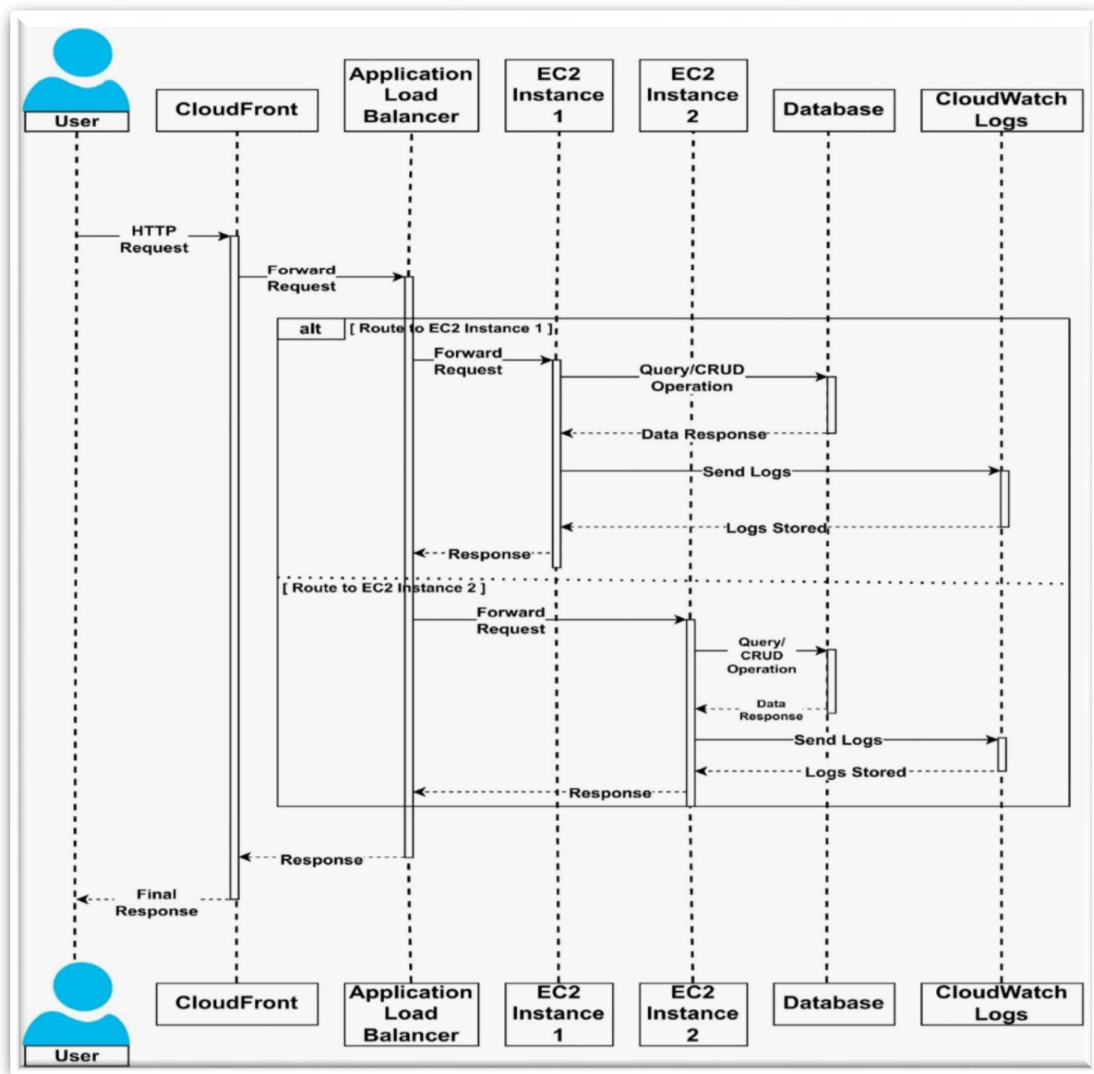


Fig: Sequence Diagram

4.5 USECASE DIAGRAM

The use-case diagram represents the interactions between the user and the system functionalities.

The primary actor in the system is the **User**, who can perform operations such as viewing, adding, updating, and deleting both books and authors. These operations represent the core CRUD functionalities of the application.

The diagram highlights how users interact with the system and what services are provided, ensuring clarity in functional requirements.

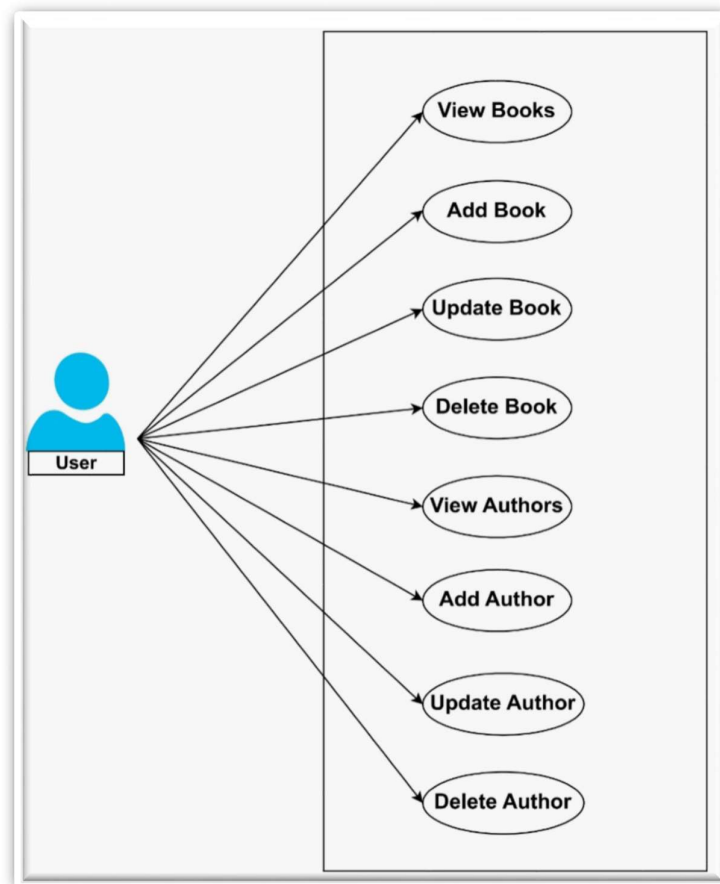


Fig: Use Case Diagram

4.7 DEPLOYMENT DIAGRAM

The deployment diagram represents the physical arrangement of system components on AWS infrastructure.

It includes user devices, Route53 for DNS resolution, CloudFront for content delivery, and the Application Load Balancer for traffic distribution. The backend is deployed on EC2 instances within an Auto Scaling Group, ensuring scalability and fault tolerance.

The database layer is implemented using Amazon RDS MySQL, while CloudWatch is used for logging and monitoring system performance. This diagram demonstrates how different AWS services are integrated to form a complete and functional system.

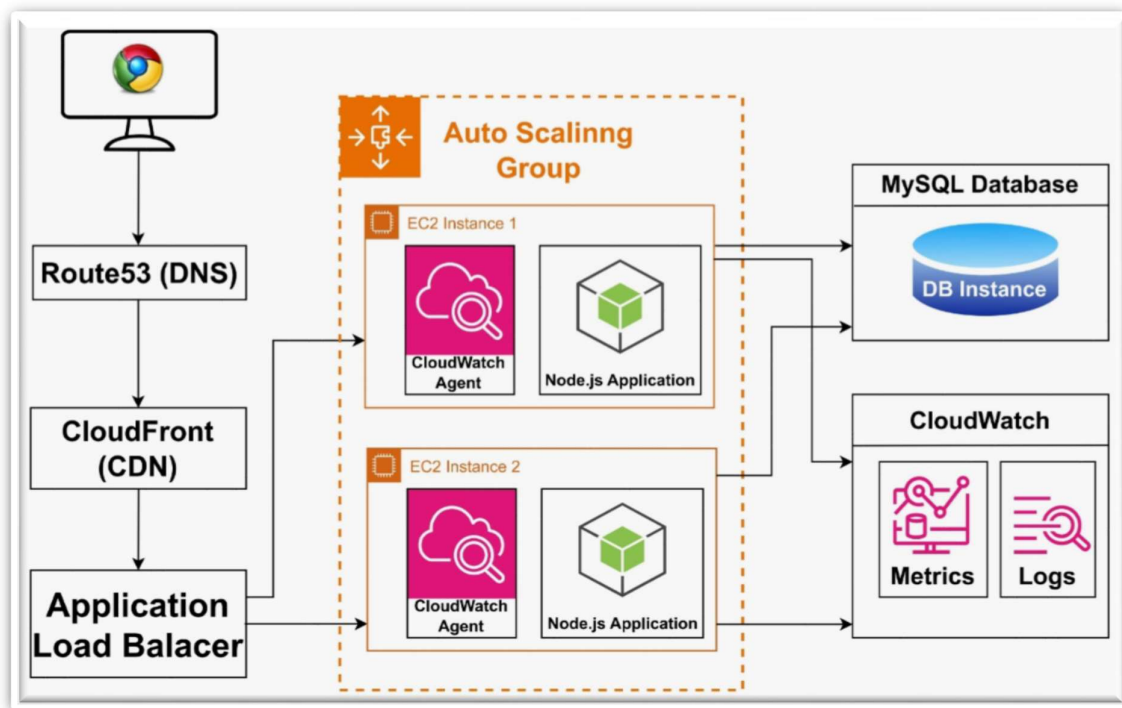


Fig: Deployment Diagram

4.8 USER INTERFACE

The user interface is developed using React.js, providing a clean and responsive experience. It includes key pages such as the Home page, Authors page, and Books page, allowing users to easily navigate the application. The interface ensures smooth interaction with the backend services and displays data dynamically in a simple and user-friendly manner.

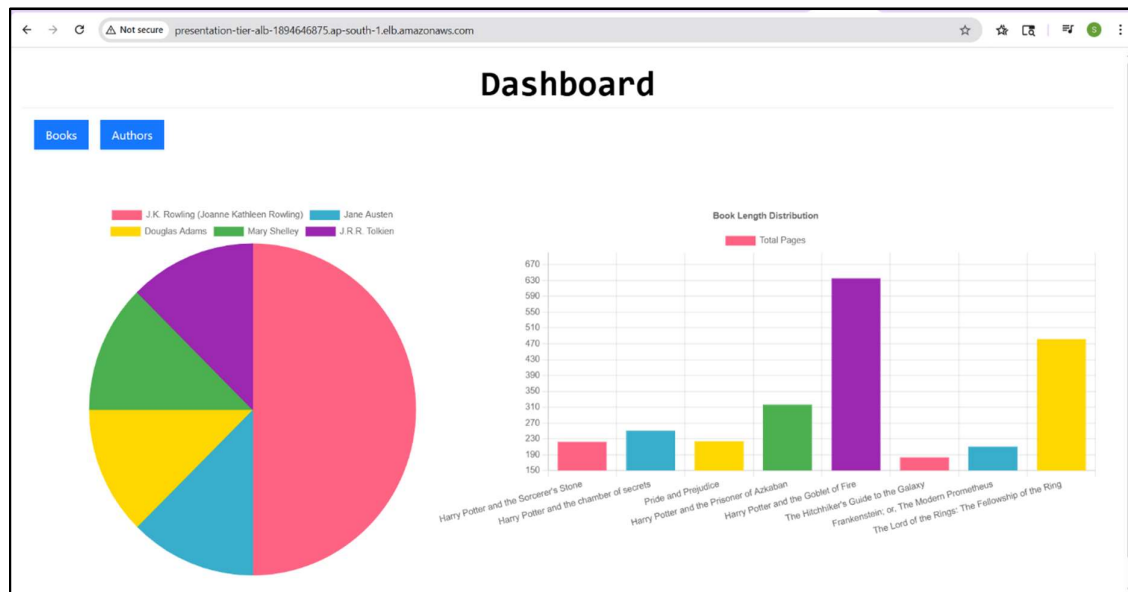


Fig 4.8.1 : Application Home Page

ID	Author	Birthday	Description	Created Date	Updated Date	Actions
1	J.K. Rowling (Joanne Kathleen Rowling)	1965-07-31T00:00:00.000Z	J.K. Rowling is a British author best known for writing the Harry Potter fantasy series. The series has won multiple awards and sold over 500 million copies, becoming the best-selling book series in history. Rowling has also written other novels, including The Casual Vacancy and the Cormoran Strike crime series under the pen name Robert Galbraith.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
3	Jane Austen	1775-12-16T00:00:00.000Z	Jane Austen was an English novelist known for her wit, social commentary, and romantic stories. Her six major novels, which explore themes of love, marriage, and money, have earned her a place as one of the greatest writers in the English language.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
4	Harper Lee	1960-07-11T00:00:00.000Z	Harper Lee was an American novelist best known for her Pulitzer Prize-winning novel To Kill a Mockingbird. The novel explores themes of racial injustice and the importance of compassion. Lee published a sequel, Go Set a Watchman, in 2015.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
5	J.R.R. Tolkien	1954-07-29T00:00:00.000Z	J.R.R. Tolkien was a British philologist and writer best known for his fantasy novels The Hobbit and The Lord of the Rings. Tolkien's works have had a profound influence on the fantasy genre and popular culture.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
6	Mary Shelley	1818-03-03T00:00:00.000Z	Mary Shelley was a British novelist, playwright, and short story writer, the daughter of Mary Wollstonecraft Godwin and the wife of poet Percy Bysshe Shelley. Frankenstein, or, The Modern Prometheus (1818) is her most famous work.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
7	Douglas Adams	1979-10-12T00:00:00.000Z	Douglas Adams was an English science fiction writer, satirist, humorist, dramatist, screenwriter, and occasional actor. He is best known for the Hitchhiker's Guide to the Galaxy comedy series, which inspired a radio comedy, several books, stage shows, comic books, a 1981 TV series, a 1984 video game, a 2005 feature film, and a 2008 sequel film.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	

Fig 4.8.2 : Authors Management Page

ID	Title	Description	Release Date	Author	Created Date	Updated Date	Actions
1	Harry Potter and the Sorcerer's Stone	On his birthday, Harry Potter discovers that he is the son of two well-known wizards, from whom he has inherited magical powers. He must attend a famous school of magic and sorcery, where he establishes a friendship with two young men who will become his companions on his adventure. During his first year at Hogwarts, he discovers that a malevolent and powerful wizard named Voldemort is in search of a philosopher's stone that prolongs the life of its owner.	1997-07-26T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
3	Harry Potter and the chamber of secrets	Harry Potter and the sophomores investigate a malevolent threat to their Hogwarts classmates, a menacing beast that hides within the castle.	1998-07-02T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
4	Pride and Prejudice	An English novel of manners by Jane Austen, first published in 1813. The story centres on the relationships among the Bennet sisters, in particular Elizabeth Bennet the middle daughter, and the wealthy Mr. Darcy. Austen satirizes the social classes of the English gentry through a witty and ironic narrative voice.	1813-01-28T00:00:00.000Z	Jane Austen	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
5	Harry Potter and the Prisoner of Azkaban	Harry's third year of studies at Hogwarts is threatened by Sirius Black's escape from Azkaban prison. Apparently, it is a dangerous wizard who was an accomplice of Lord Voldemort and who will try to take revenge on Harry Potter.	1999-07-08T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
6	Harry Potter and	Hogwarts prepares for the Triwizard Tournament, in which three schools of wizardry will compete. To	2000-07-	J.K. Rowling (Joanne	2024-05-	2024-05-	

Fig 4.8.3 : Books Management Page

IMPLEMENTATION DETAILS

The implementation of the project involves deploying a three-tier web application architecture on Amazon Web Services (AWS). The process includes configuring networking components, launching EC2 instances, setting up the RDS database, and integrating services such as Load Balancer, Auto Scaling, CloudFront, and CloudWatch.

Each step of the deployment is performed using AWS Management Console and is verified through proper configuration and testing. The following screenshots illustrate the step-by-step implementation of the system.

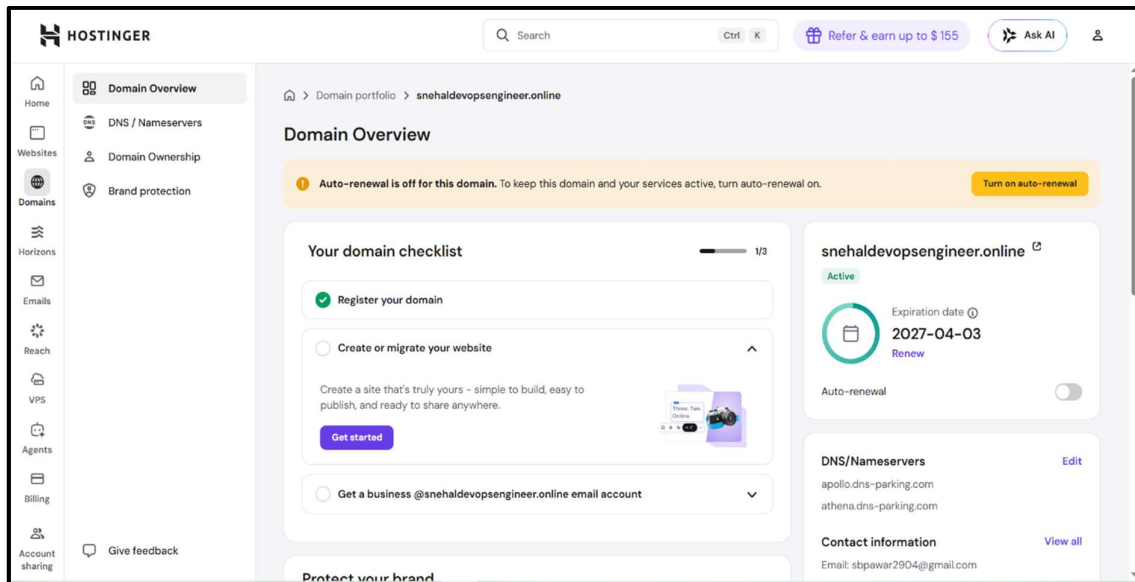


Fig 5.1: Domain Registration

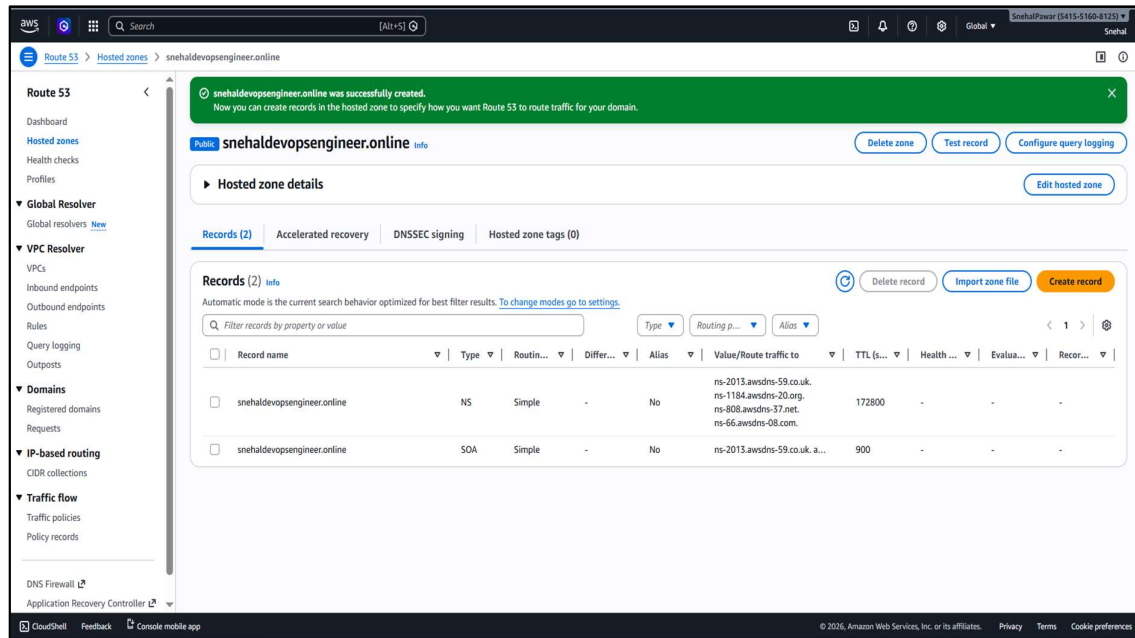


Fig 5.2: Hosted Zone Creation

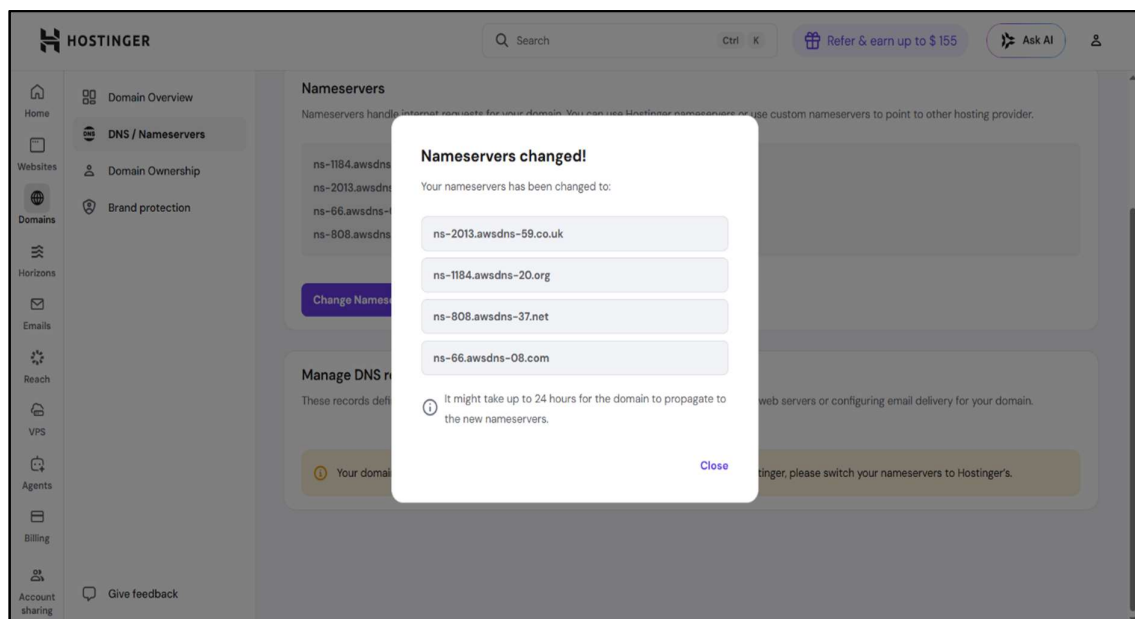


Fig 5.3: Domain Name Server Update

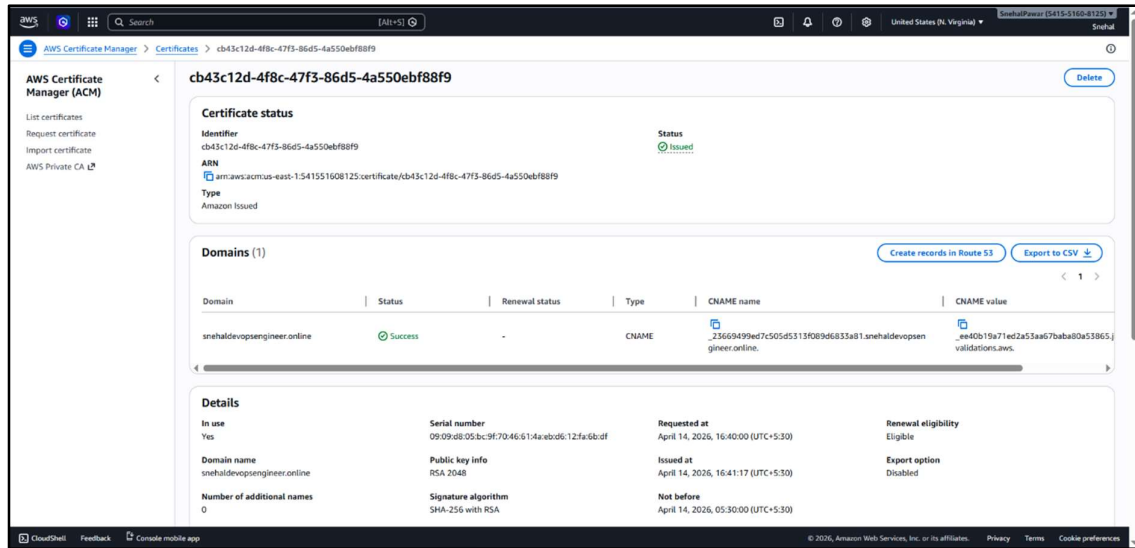


Fig 5.4: SSL Certificate Configuration (ACM)

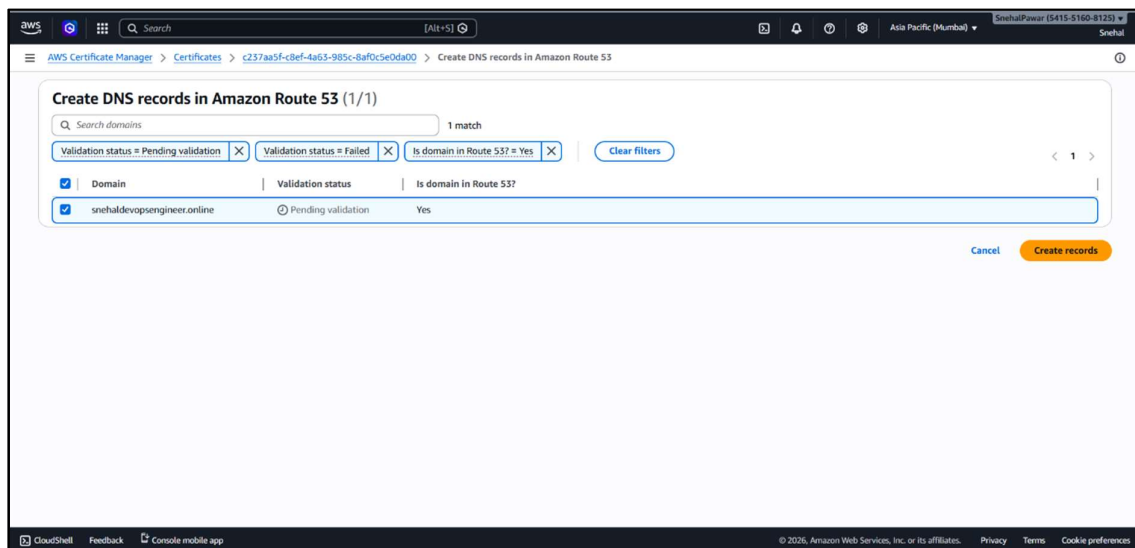


Fig 5.5: DNS Record Configuration in Route53

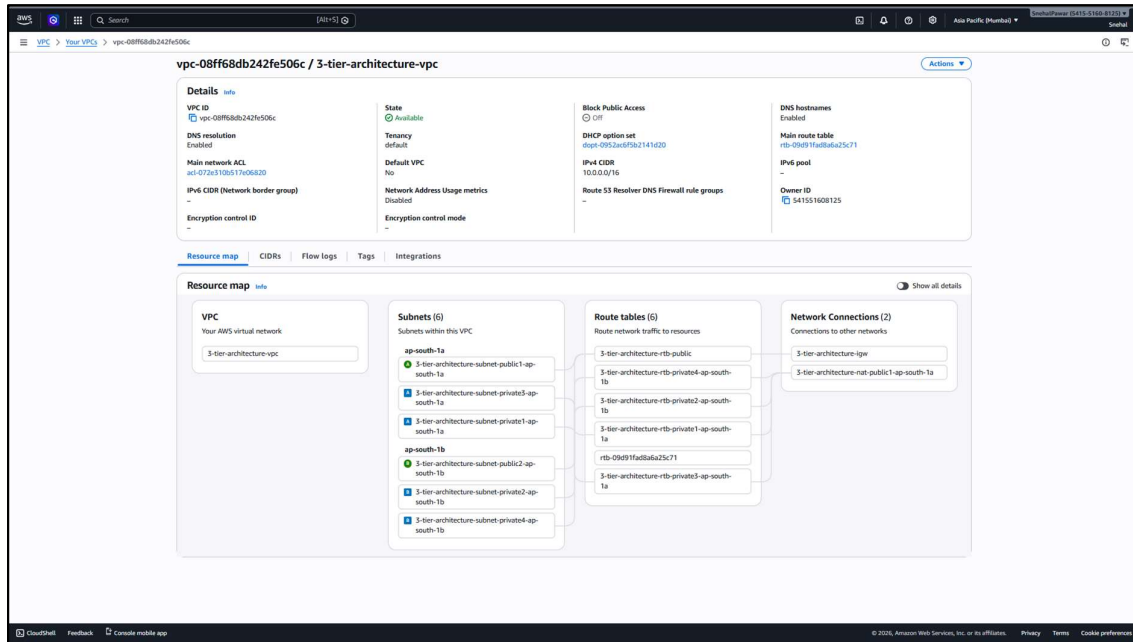


Fig 5.6: Virtual Private Cloud (VPC) Creation

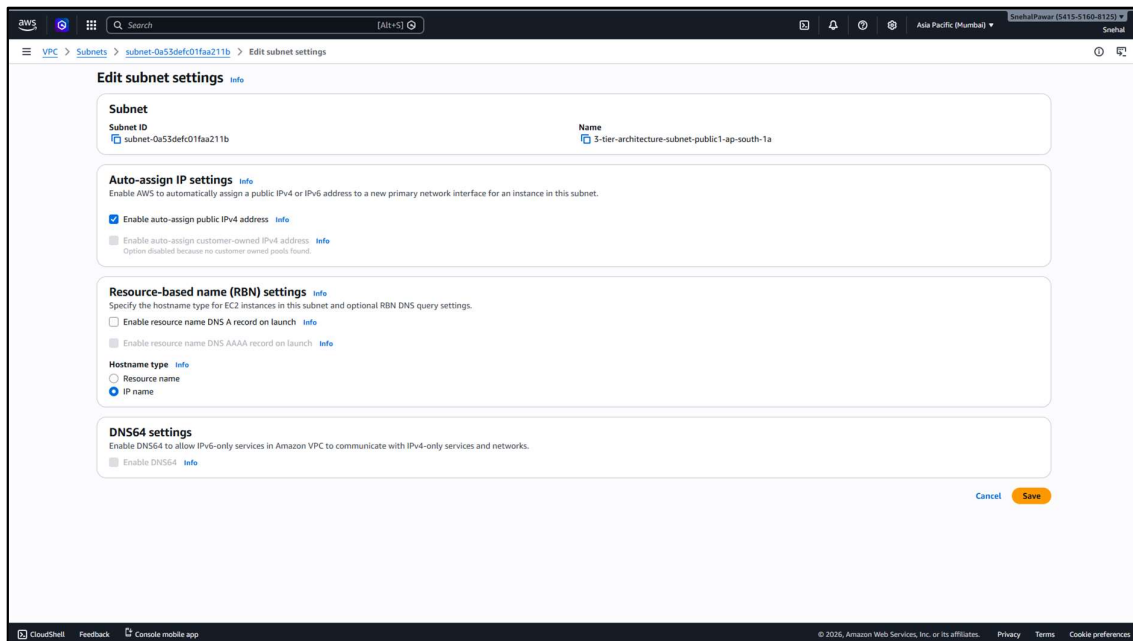


Fig 5.7: Public Subnet Configuration with Auto-Assign IP

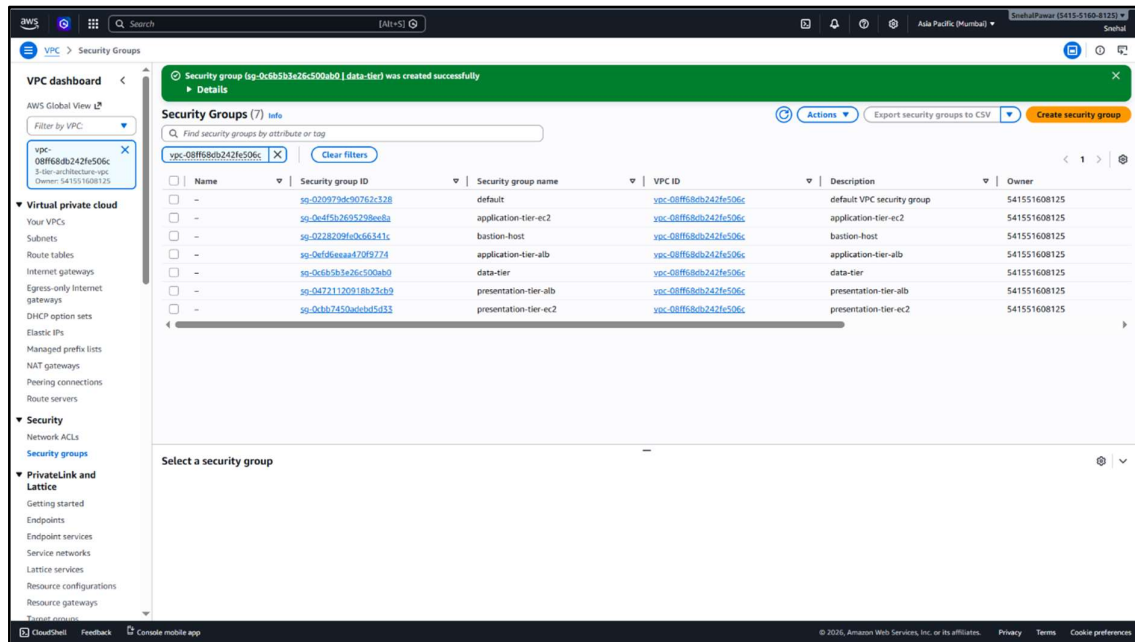


Fig 5.8: Security Group Configuration

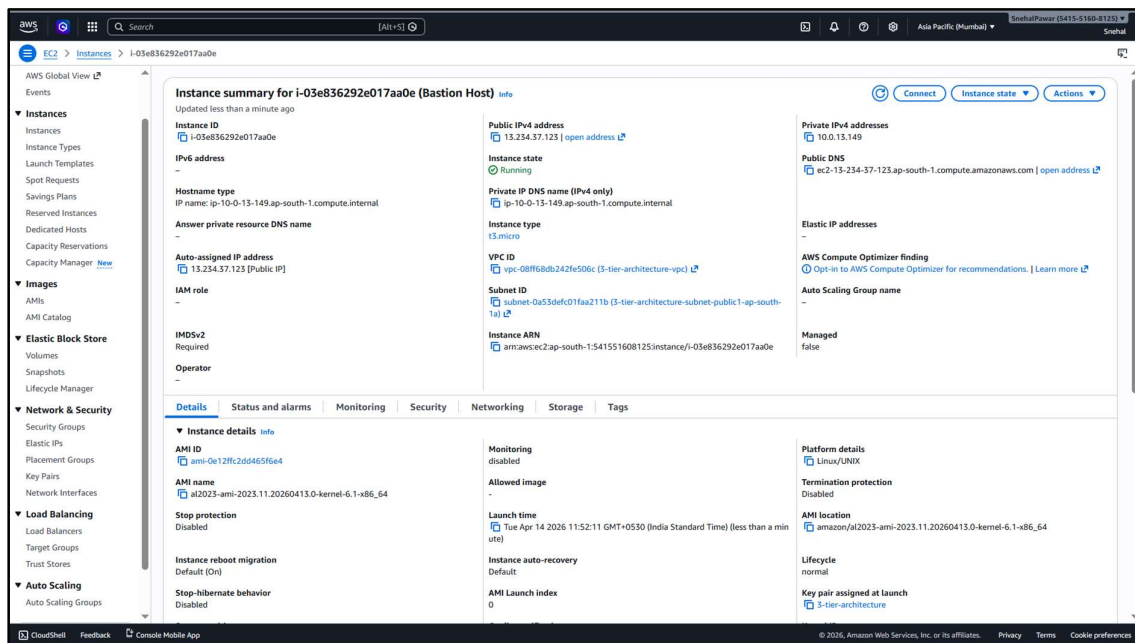


Fig 5.9: Bastion Host EC2 Instance Setup

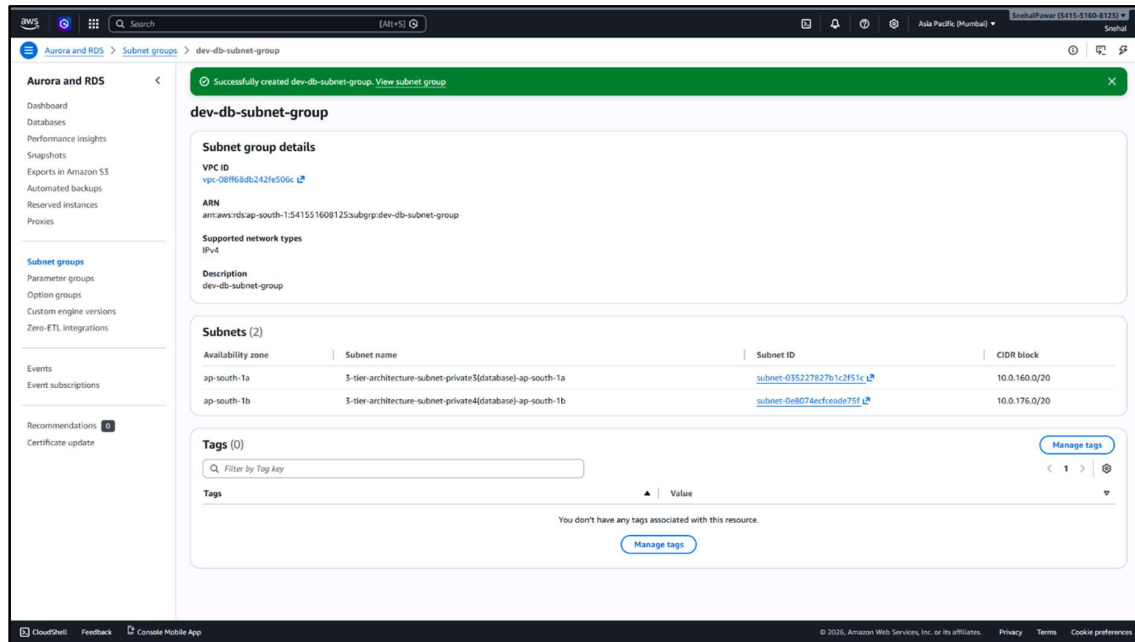


Fig 5.10: DB Subnet Group Configuration

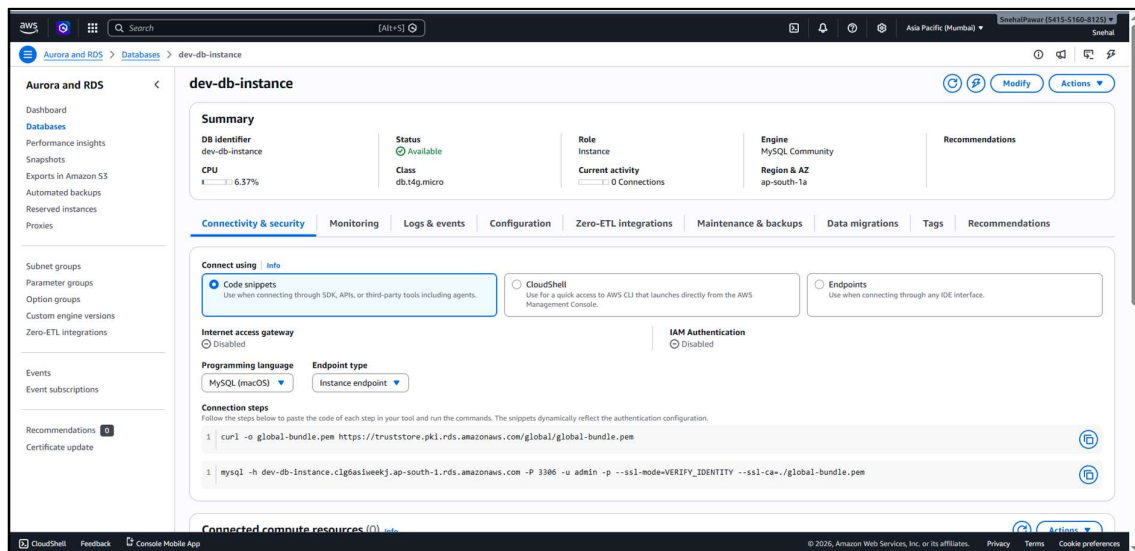


Fig 5.11: Amazon RDS Instance Creation


```
MINGW64/c/Users/Admin
Admin@DESKTOP-8SG4CSG MINGW64 ~
$ chmod 400 c:/Users/Admin/Downloads/3-tier-architecture.pem

Admin@DESKTOP-8SG4CSG MINGW64 ~
$ ssh-add ~/.ssh/3-tier-architecture.pem
Could not open a connection to your authentication agent.

Admin@DESKTOP-8SG4CSG MINGW64 ~
$ ssh -i /c/Users/Admin/Downloads/3-tier-architecture.pem -N -L 3307:dev-db-instance.clg6asiweekj.ap-south-1.rds.amazonaws.com:3306 ec2-user@ec2-13-234-37-123.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-13-234-37-123.ap-south-1.compute.amazonaws.com (13.234.37.123)' can't be established.
ED25519 key fingerprint is SHA256:ewXhN9F3Kemrg9wFGXmt9CFtEQwPjTgbZWS4BYlmeKw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-234-37-123.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
```

Fig 5.12: Secure RDS Access via SSH Tunnelling

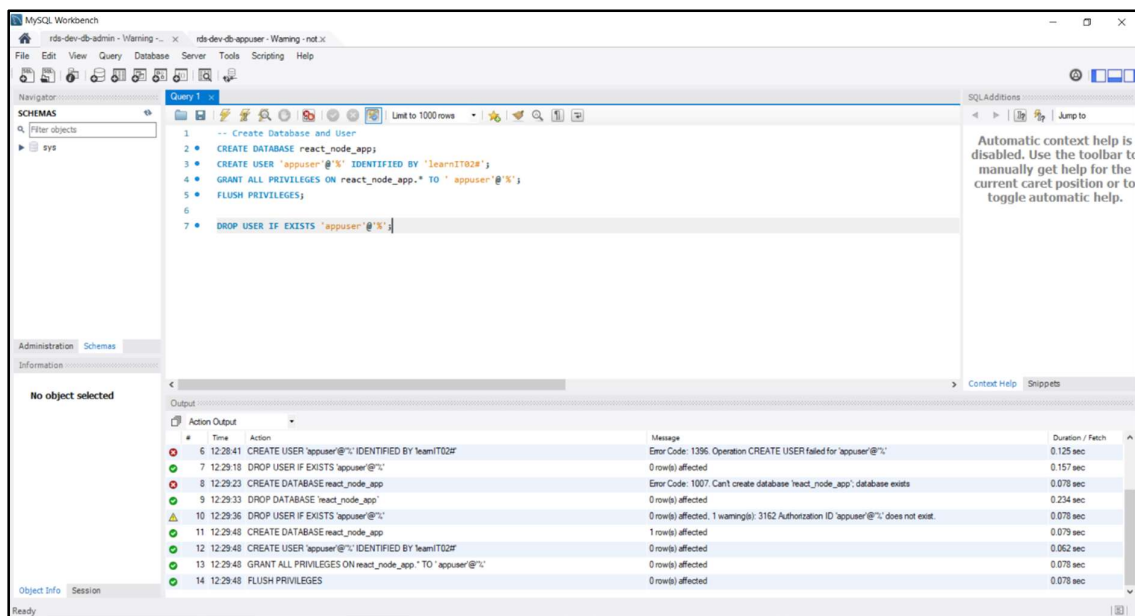


Fig 5.13: Database Access Using Admin User

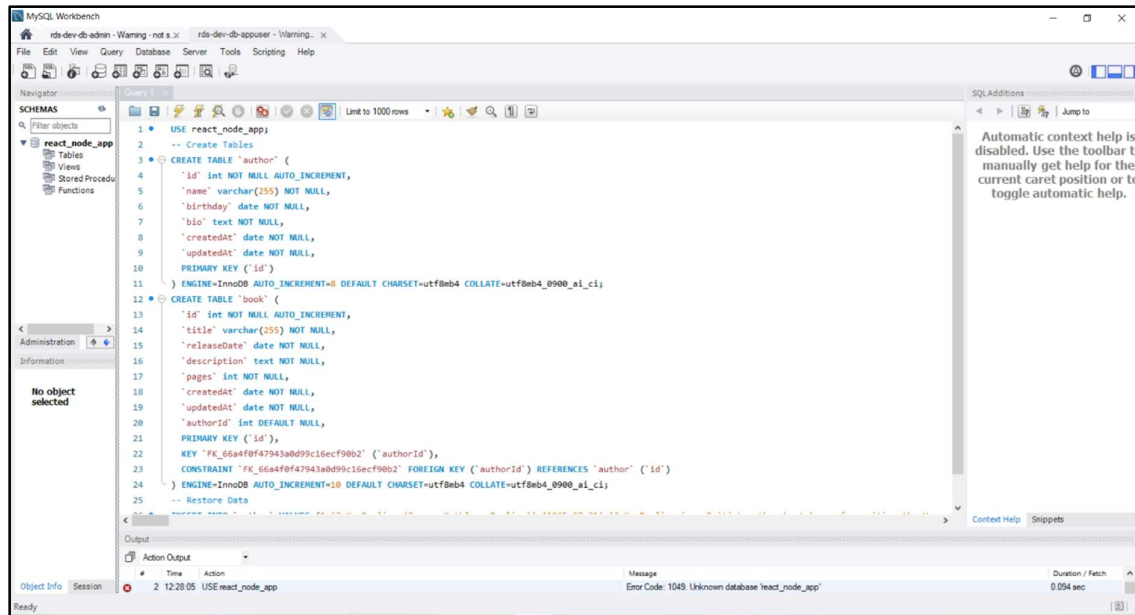


Fig 5.14: Database Access Using Application User

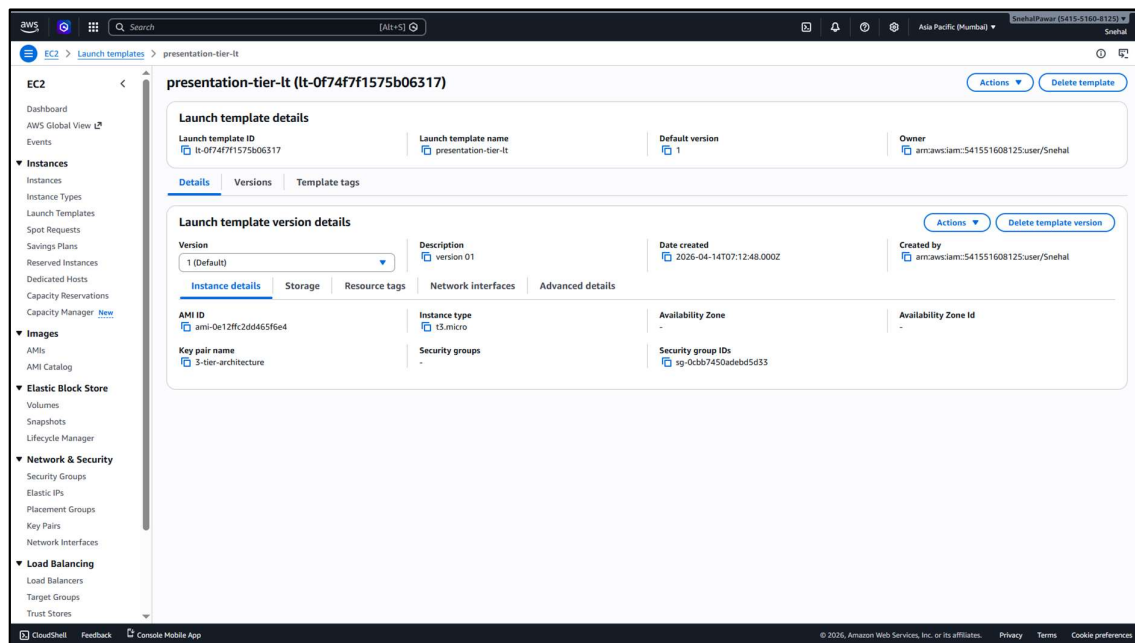


Fig 5.15: Launch Template Configuration - Presentation Tier

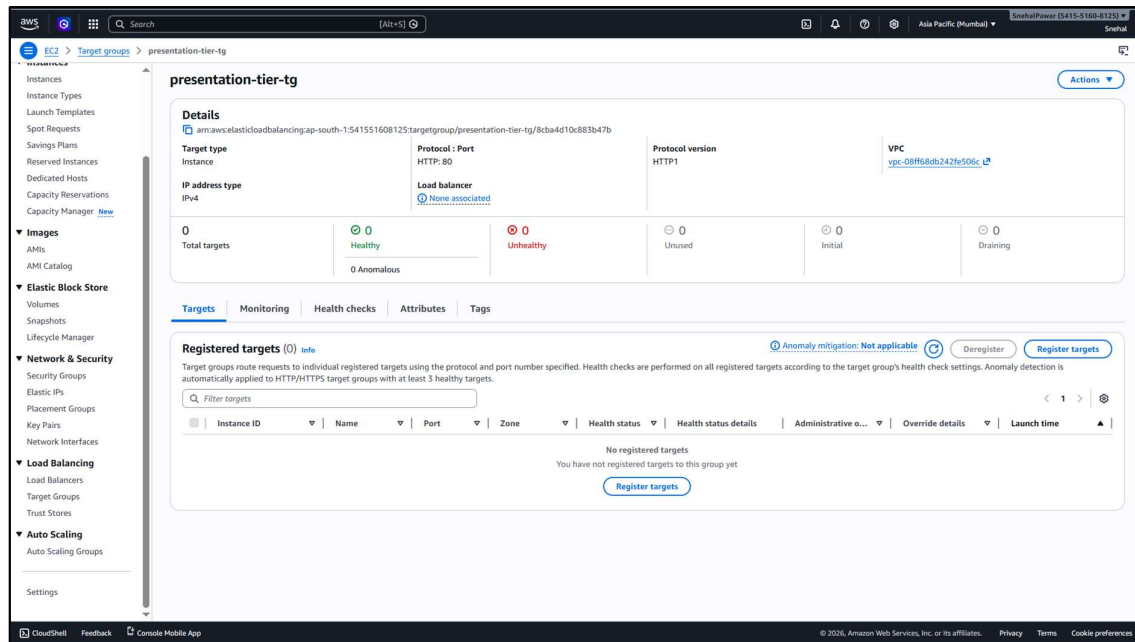


Fig 5.16: Target Group Setup - Presentation Tier

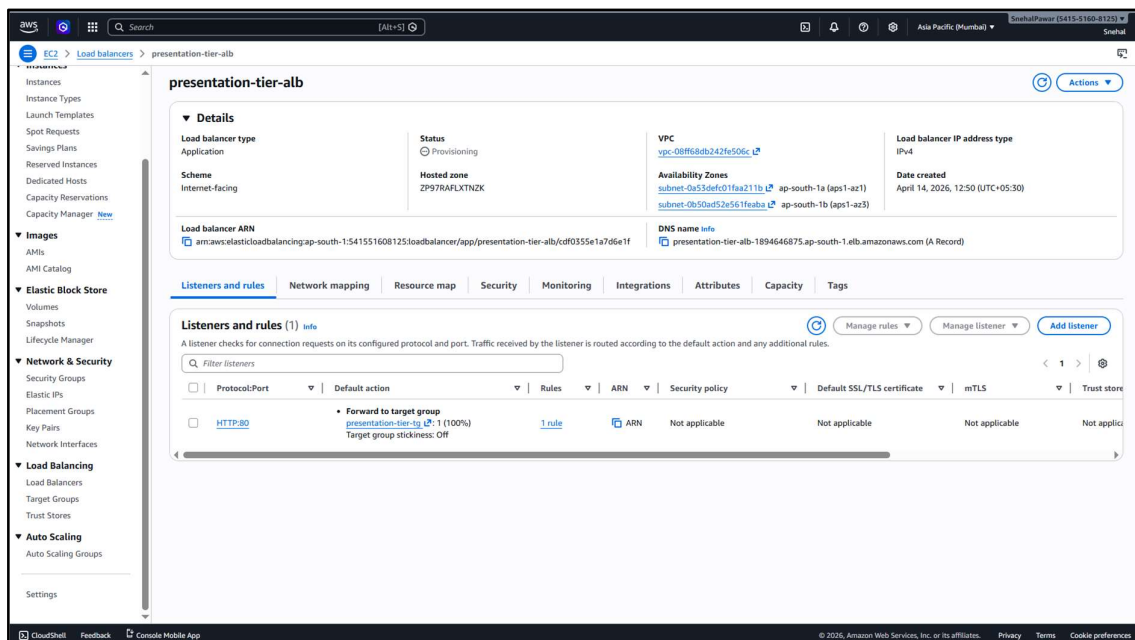


Fig 5.17: Application Load Balancer - Presentation Tier

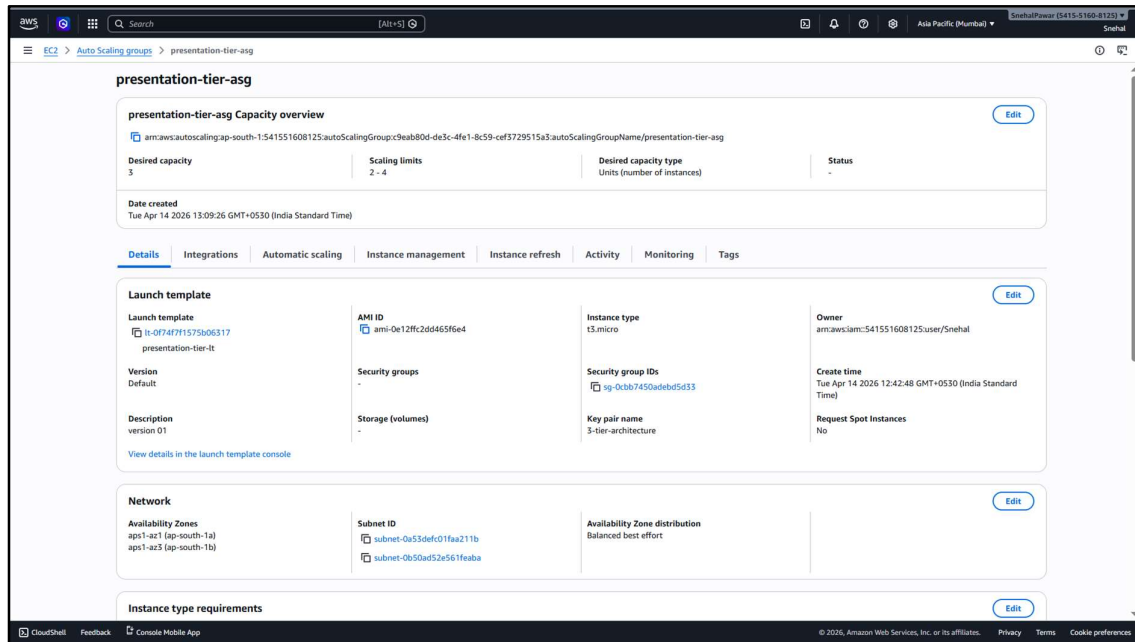


Fig 5.18: Auto Scaling Group - Presentation Tier

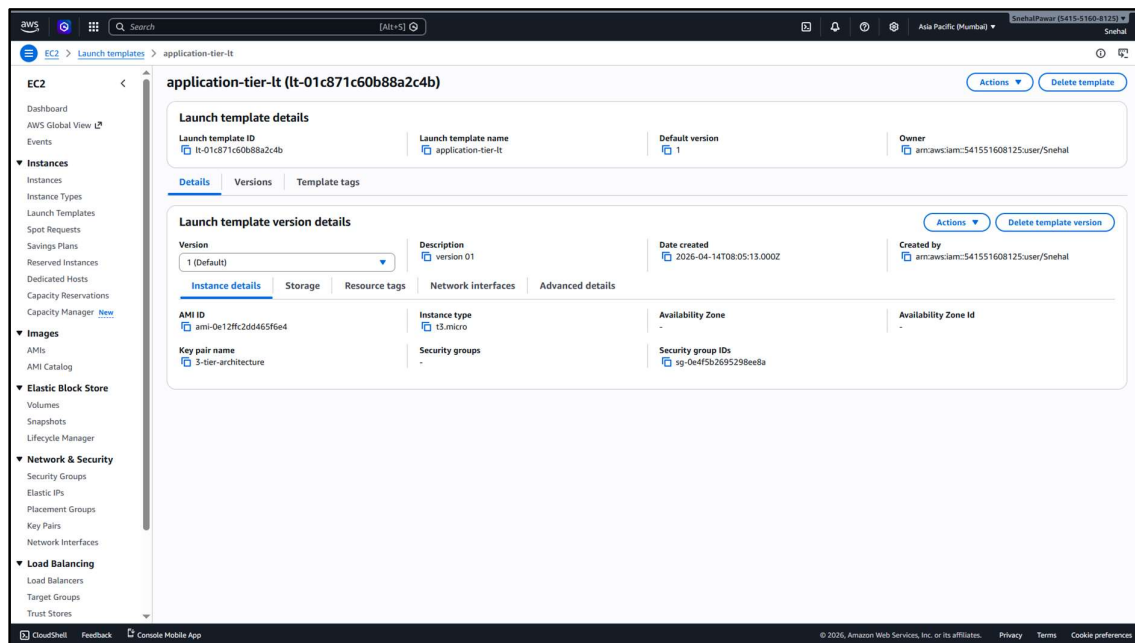


Fig 5.19: Launch Template Creation - Application Tier

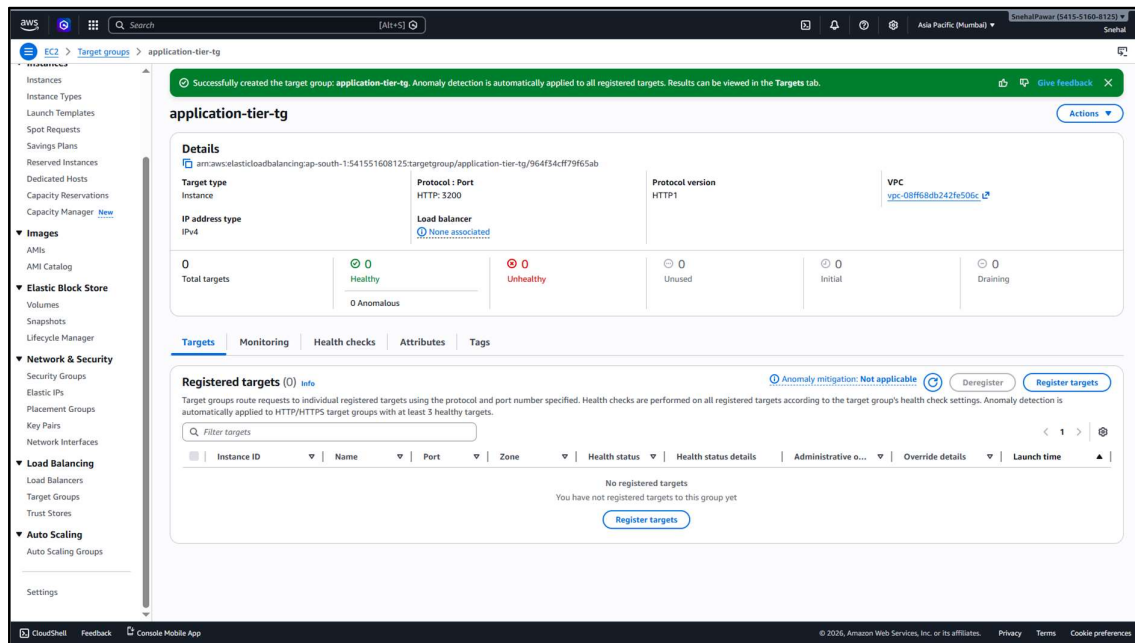


Fig 5.20: Target Group - Application Tier

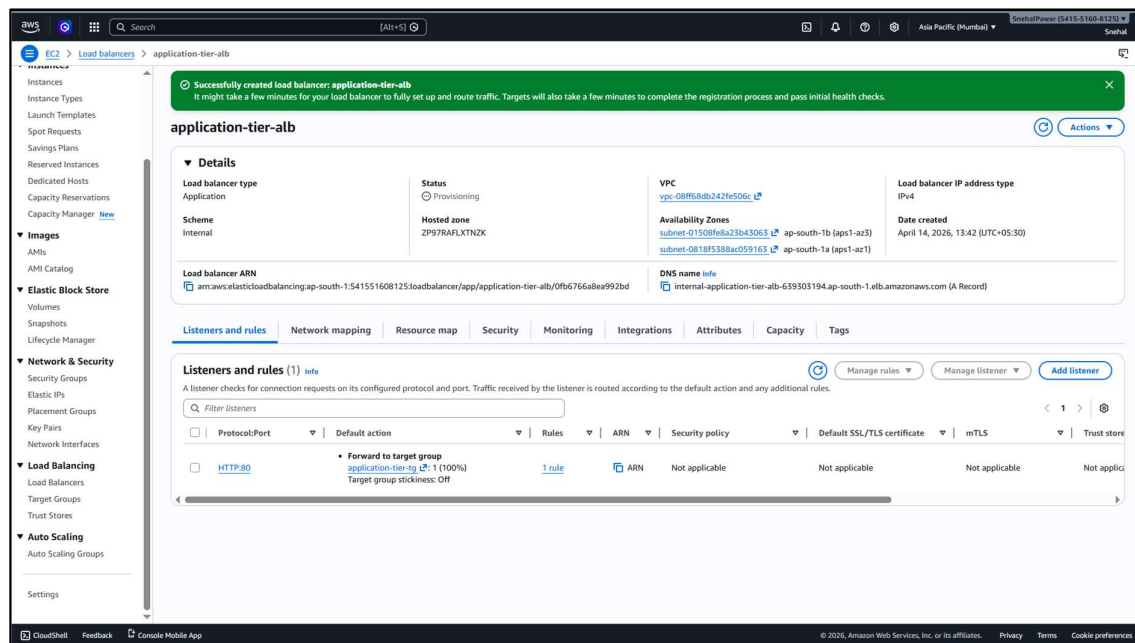


Fig 5.21: Application Load Balancer – Application Tier

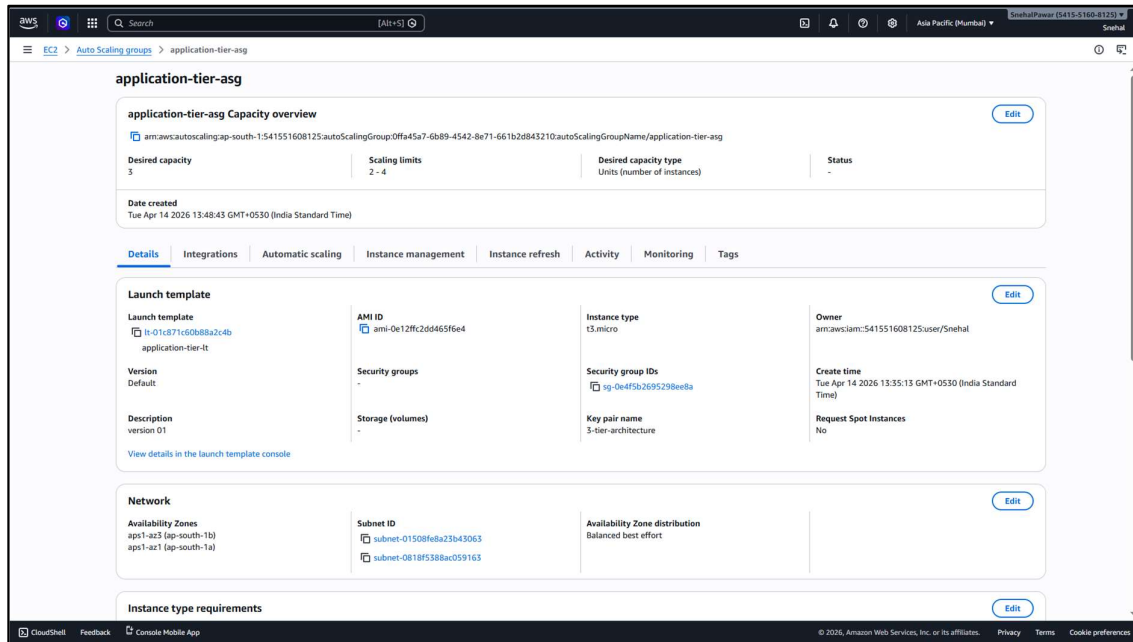


Fig 5.22: Auto Scaling Group – Application Tier

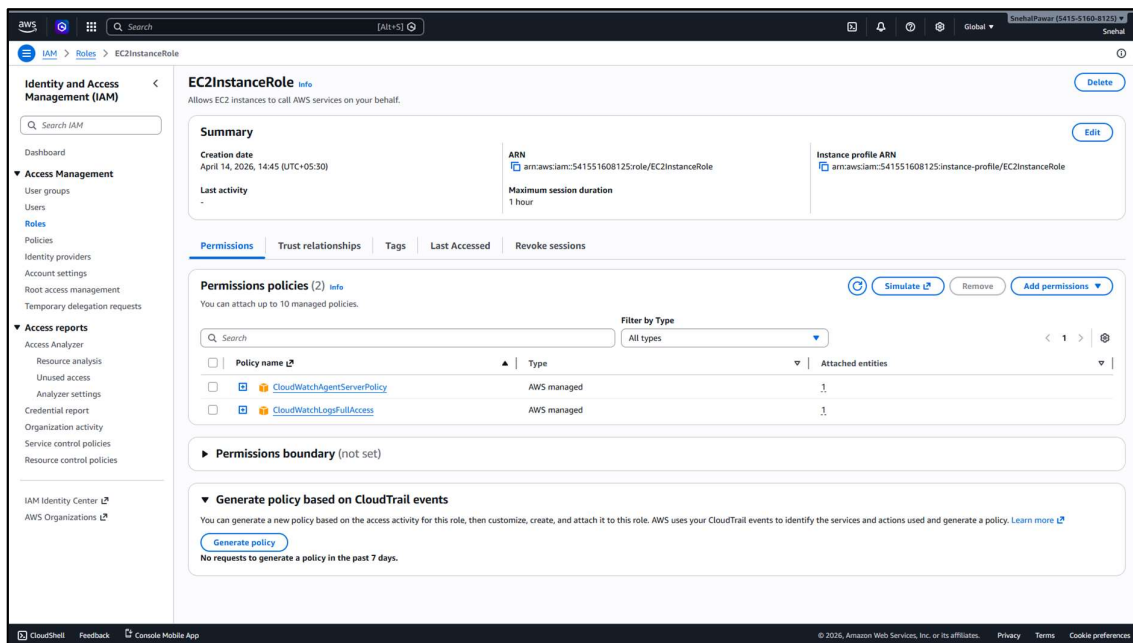


Fig 5.23: IAM Role Creation

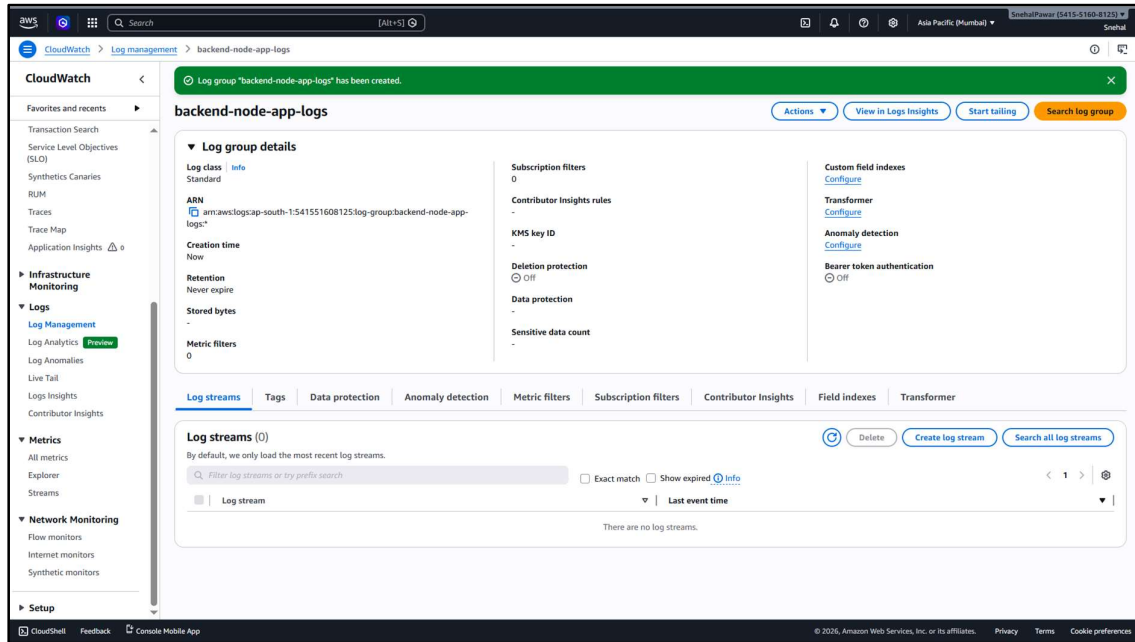


Fig 5.24: CloudWatch Log Group Setup

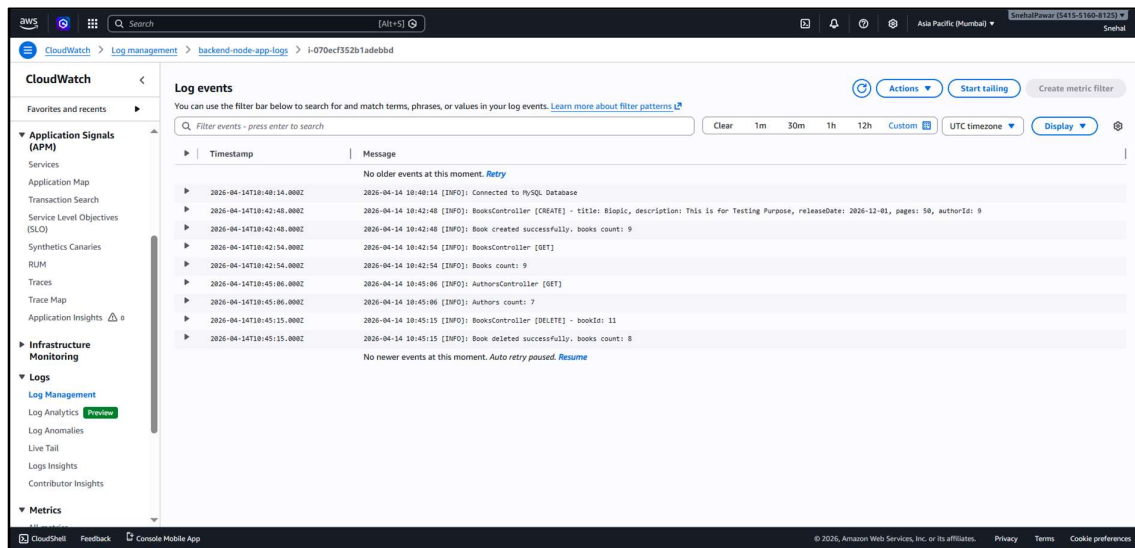


Fig 5.25: CloudWatch Logs

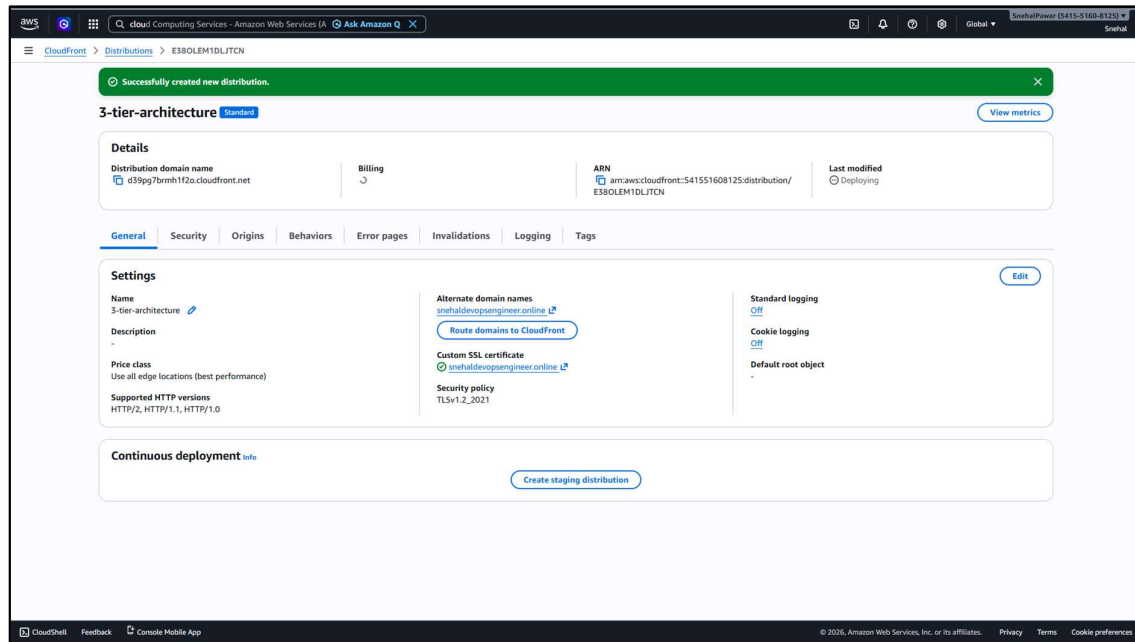


Fig 5.26: CloudFront Distribution Setup

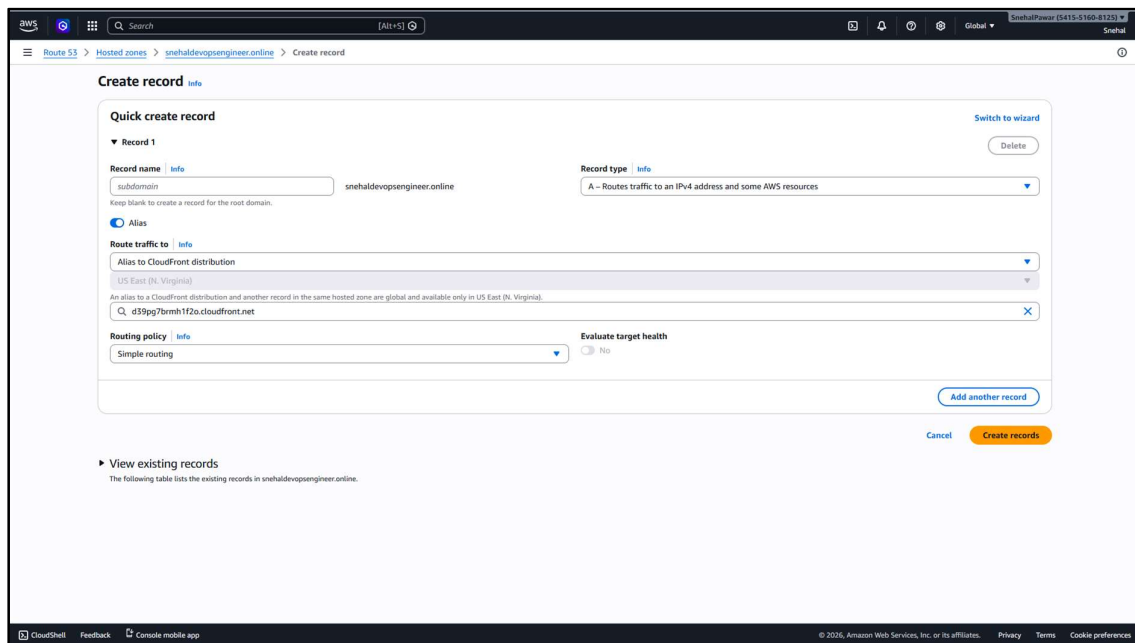


Fig 5.27: DNS Alias Record Configuration for CloudFront

SOFTWARE TESTING

Software testing in this project focuses on validating the functionality, reliability, and performance of the AWS-based three-tier architecture. The testing ensures that all cloud components such as CloudFront, Application Load Balancer, EC2 instances, RDS database, Auto Scaling Group, and CloudWatch Logs are properly integrated and working as expected.

The objective of testing is to verify that the system can handle user requests efficiently, distribute traffic correctly, maintain high availability, and generate logs for monitoring and debugging purposes.

6.1 USE CASES

The following use cases were tested to validate system functionality and cloud infrastructure behavior:

1. User Access Application

The user accesses the application using the domain or CloudFront URL. The request flows through DNS, CloudFront CDN, Application Load Balancer, and EC2 instances, and the response is displayed successfully.

2. Create Operation (Add Data)

The user adds new data through the frontend interface. The request is processed by the backend application running on EC2 and stored in the RDS database. Logs are generated in CloudWatch.

3. Read Operation (View Data)

The user requests to view stored data. The backend retrieves data from the RDS database and displays it on the frontend.

4. Update Operation

The user updates an existing record. The changes are processed by the application and updated in the database with corresponding logs generated.

5. Delete Operation

The user deletes a record. The backend removes the data from the RDS database and logs the operation.

6. Load Balancing

Multiple user requests are sent simultaneously. The Application Load Balancer distributes traffic across multiple EC2 instances to ensure efficient processing.

7. Auto Scaling

When an EC2 instance is terminated, the Auto Scaling Group automatically launches a new instance to maintain the desired capacity and ensure availability.

8. Logging and Monitoring

All backend operations generate logs which are sent to CloudWatch. Logs are organized into log groups and streams for monitoring.

9. Secure Access (HTTPS)

All HTTP requests are automatically redirected to HTTPS using CloudFront, ensuring secure communication.

6.2 TEST CASES

The following test cases were executed to validate system performance and functionality:

TID	Test Scenario	Action Performed	Expected Result	Actual Result	Status
TC01	Application Access	Open application using CloudFront URL	Application loads successfully	Application loaded successfully	Pass
TC02	Create Operation	Add new record from frontend	Data stored in RDS database	Data stored successfully	Pass
TC03	Read Operation	Fetch data from application	Data retrieved from database	Data displayed correctly	Pass
TC04	Update Operation	Modify existing record	Data updated in database	Data updated successfully	Pass
TC05	Delete Operation	Delete record from application	Record removed from database	Record deleted successfully	Pass
TC06	Load Balancing	Send multiple requests	Traffic distributed across EC2 instances	Requests distributed properly	Pass
TC07	Auto Scaling	Terminate one EC2 instance	New instance launched automatically	Instance launched successfully	Pass
TC08	Logging	Perform backend operations	Logs generated in CloudWatch	Logs visible in log group	Pass
TC09	HTTPS Redirection	Access application via HTTP	Redirected to HTTPS	Redirect successful	Pass
TC10	End-to-End Flow	Complete request cycle	Request processed & response returned	System working correctly	Pass

REPORTS

7.1 Auto Scaling Report (Presentation Tier)

The system was tested by increasing workload on the presentation tier. CloudWatch alarms monitored the CPU utilization and triggered scaling actions. As a result, additional EC2 instances were automatically launched, ensuring high availability and performance.

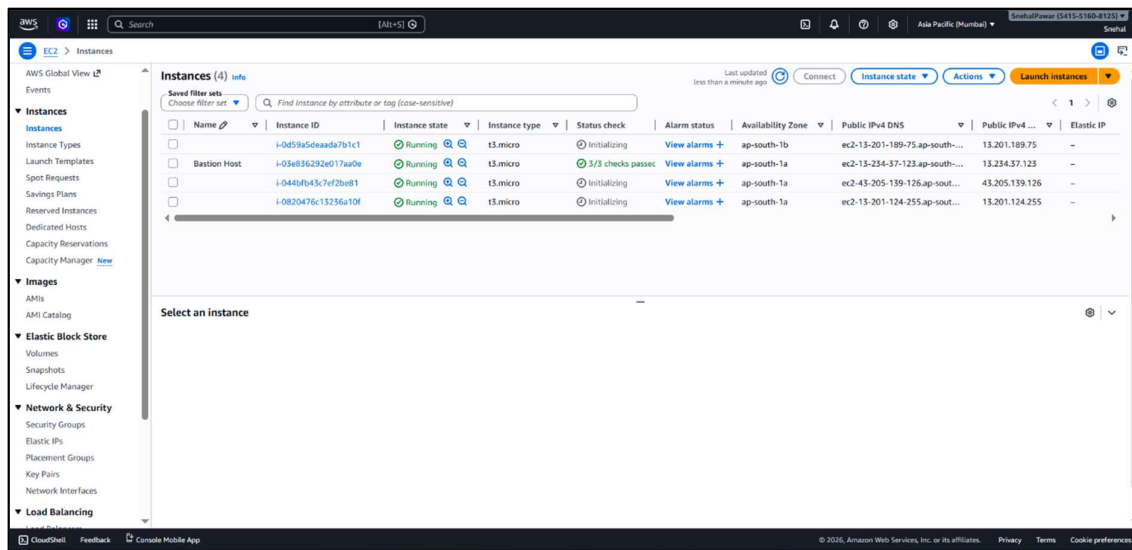


Fig 7.1.1: EC2 Instance Created by Auto Scaling Group

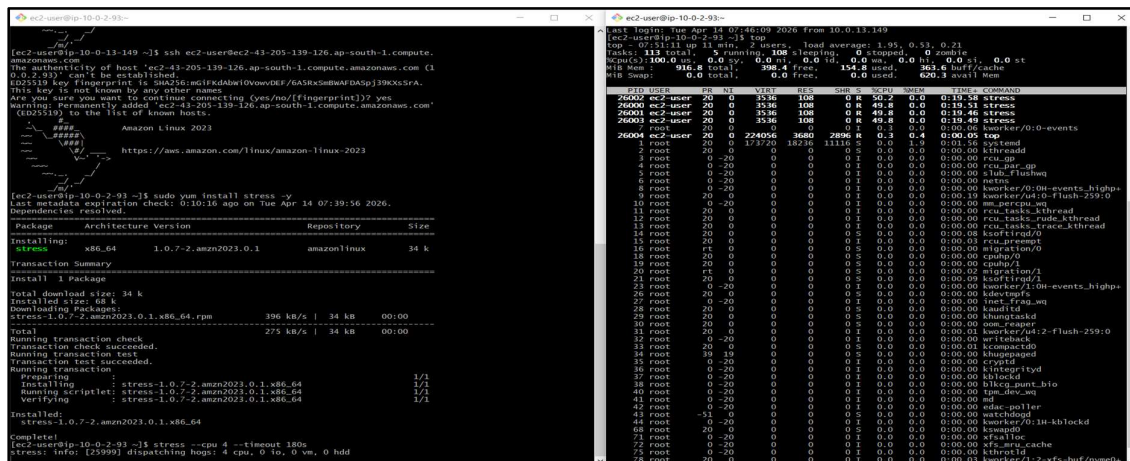


Fig 7.1.2: Load Generation

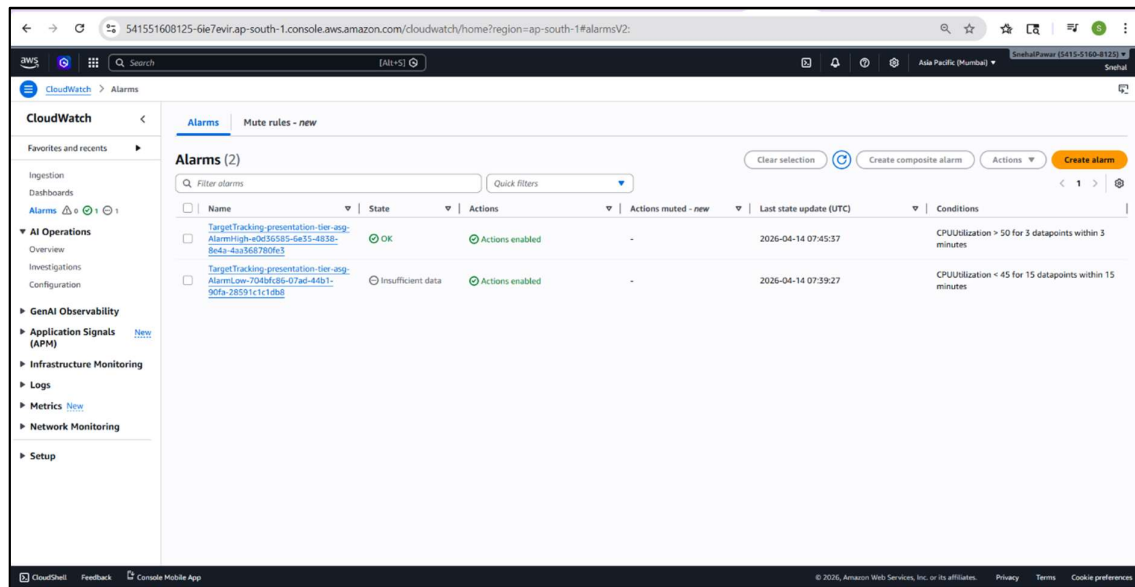


Fig 7.1.3: CloudWatch Alarm Triggered

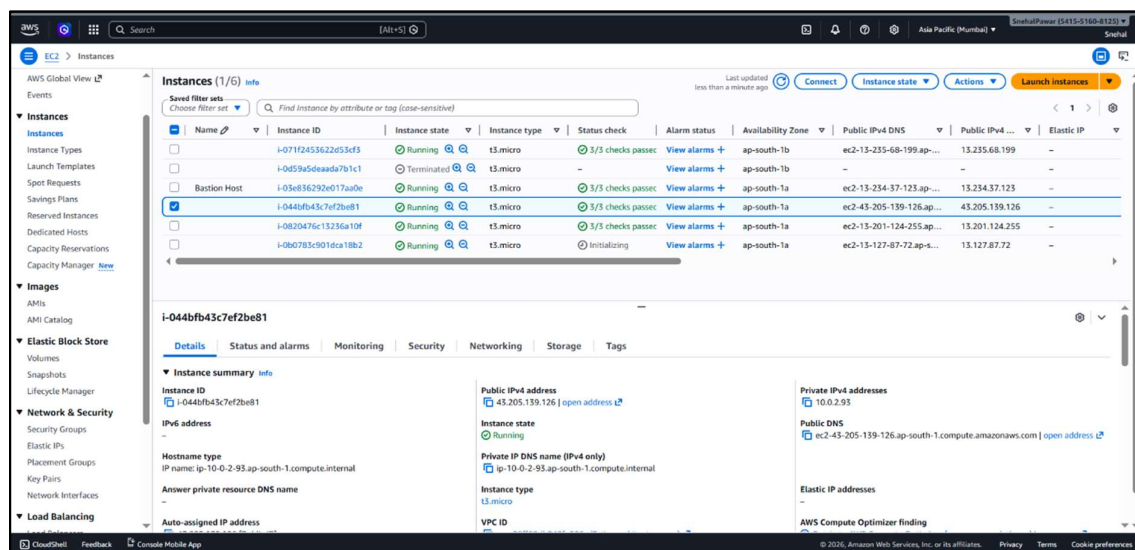
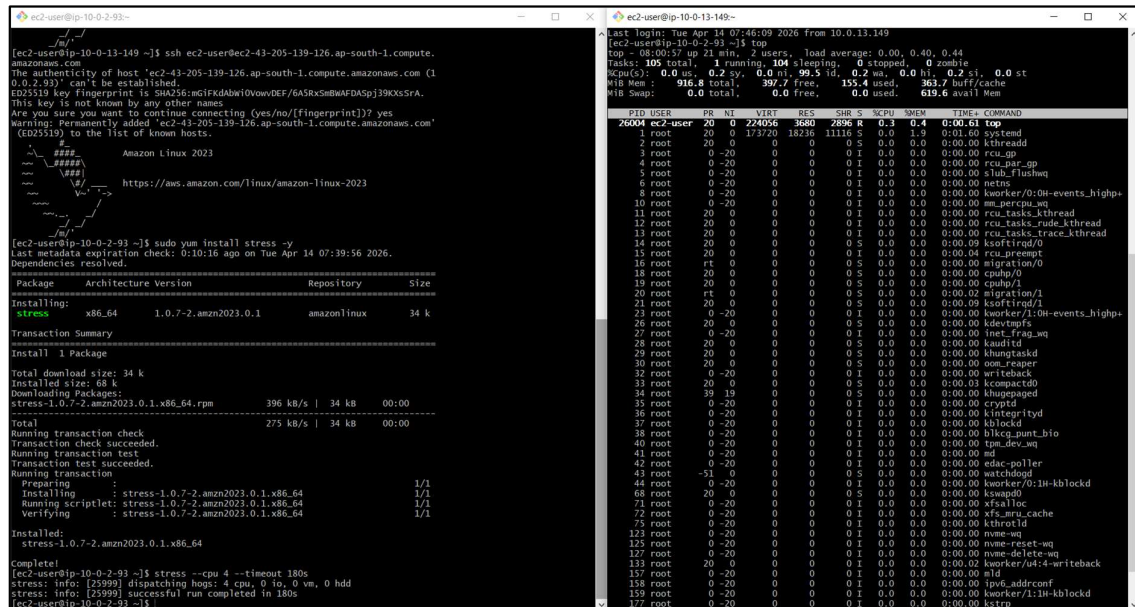


Fig 7.1.4: Auto Scaling Activity

7.2 Auto Scaling Report (Application Tier)

Similar testing was performed on the application tier by generating load. CloudWatch alarms detected increased resource usage and triggered auto scaling. New instances were launched automatically to handle backend processing efficiently.



```
ec2-user@ip-10-0-13-149:~$ ssh ec2-user@ec2-43-205-139-126.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-43-205-139-126.ap-south-1.compute.amazonaws.com (10.0.2.93)' can't be established.
202519 key fingerprint is: sha256:agIFxDbAi0vowDEF/6ASaXSmWAFDspj39KxSrA.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-43-205-139-126.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

ec2-user@ip-10-0-2-93:~$ sudo yum install stress -y
Last metadata expiration check: 0:10:16 ago on Tue Apr 14 07:39:56 2026.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
stress x86_64 1.0.7-2.amzn2023.0.1 amazonlinux 34 k
Transaction Summary
Install 1 Package
Total download size: 34 k
Installed size: 68 k
Downloading Packages:
stress-1.0.7-2.amzn2023.0.1.x86_64.rpm 396 kB/s | 34 kB 00:00
Total 275 kB/s | 34 kB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : stress-1.0.7-2.amzn2023.0.1.x86_64 1/1
Installing : stress-1.0.7-2.amzn2023.0.1.x86_64 1/1
Running scriptlet: stress-1.0.7-2.amzn2023.0.1.x86_64 1/1
Verifying : stress-1.0.7-2.amzn2023.0.1.x86_64 1/1
Installed:
stress-1.0.7-2.amzn2023.0.1.x86_64
Complete!
ec2-user@ip-10-0-2-93:~$ stress --cpu 4 --timeout 180s
stress: info: [25999] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd
stress: info: [25999] successful run completed in 180s
ec2-user@ip-10-0-2-93:~$
```

```
ec2-user@ip-10-0-13-149:~$ top
top - 08:00:57 up 21 min, 2 users, load average: 0.00, 0.40, 0.44
Tasks: 105 total, 1 running, 104 sleeping, 0 stopped, 0 zombie
Cpu(s): 0.0 us, 0.2 sy, 0.0 hi, 99.5 id, 0.2 wa, 0.0 hi, 0.2 si, 0.0 st
Mem: 916.8 total, 397.7 free, 155.4 used, 363.7 buff/cache
Mem Swap: 0.0 total, 0.0 free, 0.0 used, 619.8 avail Mem

  PID  PPID  %CPU   %MEM    VIRT   RSS   SHR   S  #TH   COMMAND
26004  ec2-user 20    0 274056 3680 2896 R  0.3  0.4  0:00.61 top
1      root    20    0 173720 18236 11116 S  0.0  1.9  0:01.60 systemd
2      root    20    0 0        0      0      S  0.0  0.0  0:00.00 kthreadd
3      root    20    0 0        0      0      S  0.0  0.0  0:00.00 rcu_gp
4      root    20    0 0        0      0      S  0.0  0.0  0:00.00 rcu_par_gp
5      root    20    0 0        0      0      S  0.0  0.0  0:00.00 slab_flushwq
6      root    20    0 0        0      0      S  0.0  0.0  0:00.00 netns
8      root    20    0 0        0      0      S  0.0  0.0  0:00.00 kworker/0:0h-events_highpri
10     root    20    0 0        0      0      S  0.0  0.0  0:00.00 mm_percpu_wq
11     root    20    0 0        0      0      S  0.0  0.0  0:00.00 rcu_tasks_kthread
12     root    20    0 0        0      0      S  0.0  0.0  0:00.00 rcu_tasks_rude_kthread
13     root    20    0 0        0      0      S  0.0  0.0  0:00.00 rcu_tasks_trace_kthread
14     root    20    0 0        0      0      S  0.0  0.0  0:00.00 ksoftirqd/0
15     root    20    0 0        0      0      S  0.0  0.0  0:00.00 rcu_preempt
16     root    rt    0 0        0      0      S  0.0  0.0  0:00.00 migration/0
18     root    20    0 0        0      0      S  0.0  0.0  0:00.00 cpuhp/0
19     root    20    0 0        0      0      S  0.0  0.0  0:00.00 cpuhp/1
20     root    rt    0 0        0      0      S  0.0  0.0  0:00.02 migration/1
21     root    20    0 0        0      0      S  0.0  0.0  0:00.00 ksoftirqd/1
23     root    20    0 0        0      0      S  0.0  0.0  0:00.00 kworker/1:0h-events_highpri
26     root    20    0 0        0      0      S  0.0  0.0  0:00.00 kdevtmpfs
27     root    20    0 0        0      0      S  0.0  0.0  0:00.00 inet_frag_wq
28     root    20    0 0        0      0      S  0.0  0.0  0:00.00 kauditd
29     root    20    0 0        0      0      S  0.0  0.0  0:00.00 khungtask
30     root    20    0 0        0      0      S  0.0  0.0  0:00.00 oom_reaper
32     root    20    0 0        0      0      S  0.0  0.0  0:00.00 writeback
33     root    20    0 0        0      0      S  0.0  0.0  0:00.00 kcompactd0
34     root    39 19 0 0        0      0      S  0.0  0.0  0:00.00 khugepaged
35     root    20    0 0        0      0      S  0.0  0.0  0:00.00 cryptd
36     root    20    0 0        0      0      S  0.0  0.0  0:00.00 kintegrityd
37     root    20    0 0        0      0      S  0.0  0.0  0:00.00 kblockd
38     root    20    0 0        0      0      S  0.0  0.0  0:00.00 blkcg_punt_bio
40     root    20    0 0        0      0      S  0.0  0.0  0:00.00 tmio_dev_wq
41     root    20    0 0        0      0      S  0.0  0.0  0:00.00 md
42     root    20    0 0        0      0      S  0.0  0.0  0:00.00 edac_poller
43     root    -51  0 0        0      0      S  0.0  0.0  0:00.00 watchdogd
44     root    20    0 0        0      0      S  0.0  0.0  0:00.00 kworker/0:1h-kblockd
68     root    20    0 0        0      0      S  0.0  0.0  0:00.00 kswapd0
71     root    20    0 0        0      0      S  0.0  0.0  0:00.00 xfsalloc
72     root    20    0 0        0      0      S  0.0  0.0  0:00.00 xfs_mru_cache
73     root    20    0 0        0      0      S  0.0  0.0  0:00.00 kthru1d
123    root    20    0 0        0      0      S  0.0  0.0  0:00.00 nvmewq
124    root    20    0 0        0      0      S  0.0  0.0  0:00.00 nvmewq
127    root    20    0 0        0      0      S  0.0  0.0  0:00.00 nvmewq
133    root    20    0 0        0      0      S  0.0  0.0  0:00.02 kworker/u4:4-writeback
157    root    20    0 0        0      0      S  0.0  0.0  0:00.00 mtd
158    root    20    0 0        0      0      S  0.0  0.0  0:00.00 ipv6_addrconf
159    root    20    0 0        0      0      S  0.0  0.0  0:00.00 kworker/1:1h-kblockd
177    root    20    0 0        0      0      S  0.0  0.0  0:00.00 kstrp
```

Fig 7.2.1: Load Generation

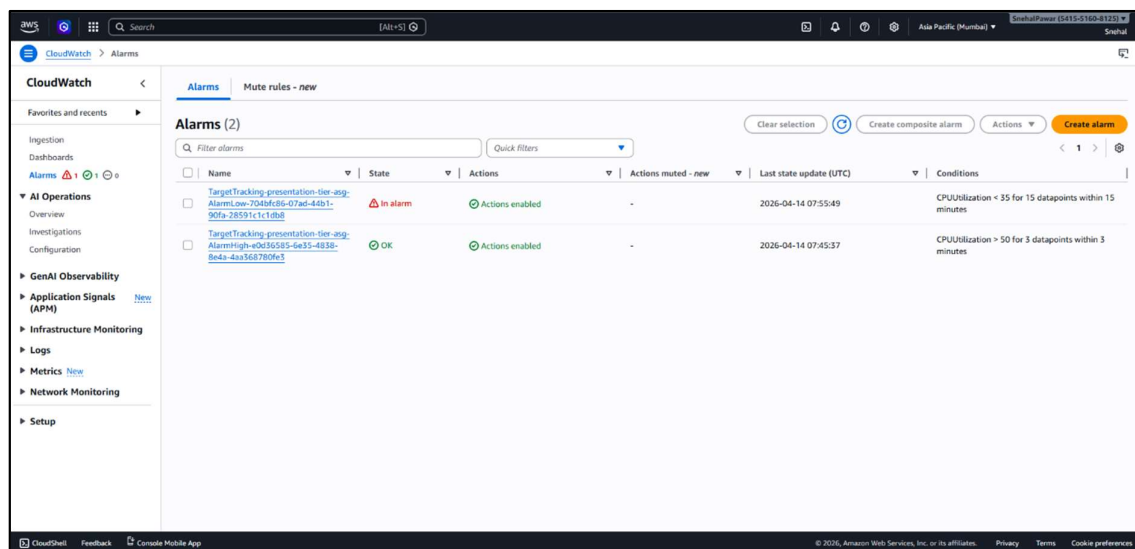


Fig 7.2.2: CloudWatch Alarm Triggered

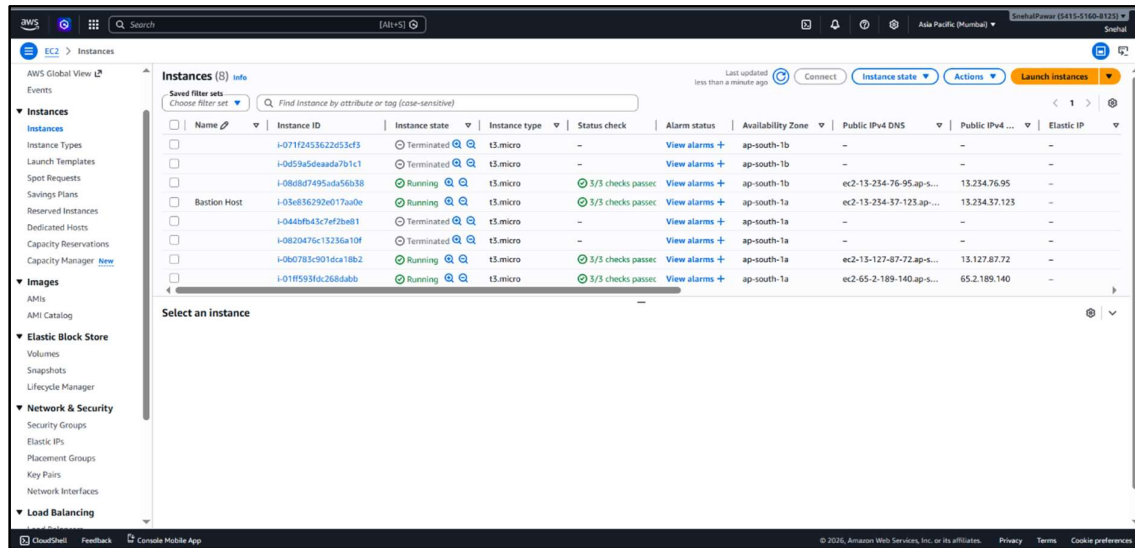


Fig 7.2.3: Auto Scaling Activity

7.3 Application Functionality Report (CRUD Operations)

The application was tested by performing CRUD (Create, Read, Update, Delete) operations. Data was successfully added, retrieved, and deleted from the system, confirming proper integration between frontend, backend, and database.

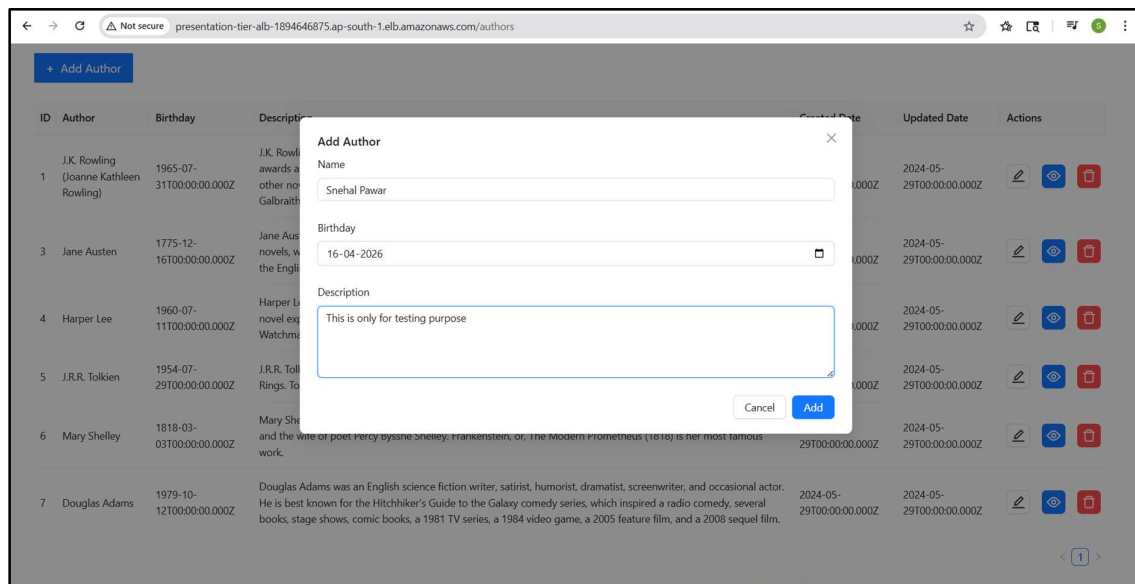


Fig 7.3.1: Adding a New Author

MANAGE AUTHORS

[+ Add Author](#)

ID	Author	Birthday	Description	Created Date	Updated Date	Actions
1	J.K. Rowling (Joanne Kathleen Rowling)	1965-07-31T00:00:00.000Z	J.K. Rowling is a British author best known for writing the Harry Potter fantasy series. The series has won multiple awards and sold over 500 million copies, becoming the best-selling book series in history. Rowling has also written other novels, including The Casual Vacancy and the Cormoran Strike crime series under the pen name Robert Galbraith.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
3	Jane Austen	1775-12-16T00:00:00.000Z	Jane Austen was an English novelist known for her wit, social commentary, and romantic stories. Her six major novels, which explore themes of love, marriage, and money, have earned her a place as one of the greatest writers in the English language.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
4	Harper Lee	1960-07-11T00:00:00.000Z	Harper Lee was an American novelist best known for her Pulitzer Prize-winning novel To Kill a Mockingbird. The novel explores themes of racial injustice and the importance of compassion. Lee published a sequel, Go Set a Watchman, in 2015.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
5	J.R.R. Tolkien	1954-07-29T00:00:00.000Z	J.R.R. Tolkien was a British philologist and writer best known for his fantasy novels The Hobbit and The Lord of the Rings. Tolkien's works have had a profound influence on the fantasy genre and popular culture.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
6	Mary Shelley	1818-03-03T00:00:00.000Z	Mary Shelley was a British novelist, playwright, and short story writer, the daughter of Mary Wollstonecraft Godwin and the wife of poet Percy Bysshe Shelley. Frankenstein, or, The Modern Prometheus (1818) is her most famous work.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
7	Douglas Adams	1979-10-12T00:00:00.000Z	Douglas Adams was an English science fiction writer, satirist, humorist, dramatist, screenwriter, and occasional actor. He is best known for the Hitchhiker's Guide to the Galaxy comedy series, which inspired a radio comedy, several books, stage shows, comic books, a 1981 TV series, a 1984 video game, a 2005 feature film, and a 2008 sequel film.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
8	Snehal Pawar	2026-04-16T00:00:00.000Z	This is only for testing purpose	2026-04-14T00:00:00.000Z	2026-04-14T00:00:00.000Z	

[1](#)

Fig 7.3.2: Author Data Successfully Added

MANAGE BOOKS

[+ Add Book](#)

Add Book

Title:

Release Date:

Description:

Pages:

Author:

[Cancel](#) [Add](#)

ID	Title	Description	Created Date	Updated Date	Actions
1	Harry Potter and the Sorcerer's Stone	On his birthday, Harry Potter discovers the powers. He must attend a famous school and will become his companions on his adventure. A powerful wizard named Voldemort is in search of him.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
3	Harry Potter and the chamber of secrets	Harry Potter and the sophomores investigate the hidden dangers within the castle.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
4	Pride and Prejudice	An English novel of manners by Jane Austen. It follows the Bennet sisters, in particular Elizabeth Bennet, as they navigate the social classes of the English gentry through a series of misunderstandings.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
5	Harry Potter and the Prisoner of Azkaban	Harry's third year of studies at Hogwarts is filled with danger as he uncovers the truth about a dangerous wizard who was an accomplice of Voldemort.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
6	Harry Potter and the Goblet of Fire	Hogwarts prepares for the Triwizard Tournament. Harry Potter is chosen to participate in the tournament, which is more dangerous than the previous ones.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	
7	The Hitchhiker's Guide to the Galaxy	A comic science fiction comedy series created by Douglas Adams. The story follows the comedic misadventures of Arthur Dent, a hapless Englishman, following the destruction of the Earth by the Vogons, a race of unpleasant bureaucratic aliens.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	

Fig 7.3.3: Adding a New Book

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ID	Title	Description	Release Date	Author	Created Date	Updated Date	Actions
1	Harry Potter and the Sorcerer's Stone	On his birthday, Harry Potter discovers that he is the son of two well-known wizards, from whom he has inherited magical powers. He must attend a famous school of magic and sorcery, where he establishes a friendship with two young men who will become his companions on his adventure. During his first year at Hogwarts, he discovers that a malevolent and powerful wizard named Voldemort is in search of a philosopher's stone that prolongs the life of its owner.	1997-07-26T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
3	Harry Potter and the chamber of secrets	Harry Potter and the sophomores investigate a malevolent threat to their Hogwarts classmates, a menacing beast that hides within the castle.	1998-07-02T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
4	Pride and Prejudice	An English novel of manners by Jane Austen, first published in 1813. The story centres on the relationships among the Bennet sisters, in particular Elizabeth Bennet the middle daughter, and the wealthy Mr. Darcy. Austen satirizes the social classes of the English gentry through a witty and ironic narrative voice.	1813-01-28T00:00:00.000Z	Jane Austen	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
5	Harry Potter and the Prisoner of Azkaban	Harry's third year of studies at Hogwarts is threatened by Sirius Black's escape from Azkaban prison. Apparently, it is a dangerous wizard who was an accomplice of Lord Voldemort and who will try to take revenge on Harry Potter.	1999-07-08T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
6	Harry Potter and the Goblet of Fire	Hogwarts prepares for the Triwizard Tournament, in which three schools of wizardry will compete. To everyone's surprise, Harry Potter is chosen to participate in the competition, in which he must fight dragons, enter the water and face his greatest fears.	2000-07-08T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
7	The Hitchhiker's Guide to the Galaxy	A comic science fiction comedy series created by Douglas Adams. The story follows the comedic misadventures of Arthur Dent, a hapless Englishman, following the destruction of the Earth by the Vogons, a race of unpleasant bureaucratic aliens. Arthur escapes with his friend Ford Prefect, who reveals himself to be an undercover researcher for the titular Hitchhiker's Guide to the Galaxy, a galactic encyclopedia containing information about anything and everything.	1979-10-12T00:00:00.000Z	Douglas Adams	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
8	Frankenstein; or, The Modern Prometheus	A Gothic novel by Mary Shelley that tells the story of Victor Frankenstein, a young scientist who creates a grotesque creature in an unorthodox scientific experiment. Frankenstein is horrified by his creation and abandons it, but the creature seeks revenge. The novel explores themes of scientific responsibility, creation versus destruction, and the nature of good and evil.	1818-03-03T00:00:00.000Z	Mary Shelley	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
9	The Lord of the Rings: The Fellowship of the Ring	The first book in J.R.R. Tolkien's epic fantasy trilogy, The Lord of the Rings. The Fellowship of the Ring follows hobbit Frodo Baggins as he inherits the One Ring, an evil artifact of power created by the Dark Lord Sauron. Frodo embarks on a quest to destroy the Ring in the fires of Mount Doom, accompanied by a fellowship of eight companions.	1954-07-29T00:00:00.000Z	J.R.R. Tolkien	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
10	Biopic	This is only for testing purpose	2026-12-01T00:00:00.000Z	Snehal Pawar	2026-04-14T00:00:00.000Z	2026-04-14T00:00:00.000Z	  

Fig 7.3.4: Book Data Successfully Added

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5	Harry Potter and the Prisoner of Azkaban	Harry's third year of studies at Hogwarts is threatened by Sirius Black's escape from Azkaban prison. Apparently, it is a dangerous wizard who was an accomplice of Lord Voldemort and who will try to take revenge on Harry Potter.	1999-07-08T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
6	Harry Potter and the Goblet of Fire	Hogwarts prepares for the Triwizard Tournament, everyone's surprise, Harry Potter is chosen to participate, in which he must fight dragons, enter the water and face his greatest fears.			2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
7	The Hitchhiker's Guide to the Galaxy	A comic science fiction comedy series created by Douglas Adams. The story follows the comedic misadventures of Arthur Dent, a hapless Englishman, following the destruction of the Earth by the Vogons, a race of unpleasant bureaucratic aliens. Arthur escapes with his friend Ford Prefect, who reveals himself to be an undercover researcher for the titular Hitchhiker's Guide to the Galaxy, a galactic encyclopedia containing information about anything and everything.	1979-10-12T00:00:00.000Z	Douglas Adams	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
8	Frankenstein; or, The Modern Prometheus	A Gothic novel by Mary Shelley that tells the story of Victor Frankenstein, a young scientist who creates a grotesque creature in an unorthodox scientific experiment. Frankenstein is horrified by his creation and abandons it, but the creature seeks revenge. The novel explores themes of scientific responsibility, creation versus destruction, and the nature of good and evil.	1818-03-03T00:00:00.000Z	Mary Shelley	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
9	The Lord of the Rings: The Fellowship of the Ring	The first book in J.R.R. Tolkien's epic fantasy trilogy, The Lord of the Rings. The Fellowship of the Ring follows hobbit Frodo Baggins as he inherits the One Ring, an evil artifact of power created by the Dark Lord Sauron. Frodo embarks on a quest to destroy the Ring in the fires of Mount Doom, accompanied by a fellowship of eight companions.	1954-07-29T00:00:00.000Z	J.R.R. Tolkien	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
10	Biopic	This is only for testing purpose	2026-12-01T00:00:00.000Z	Snehal Pawar	2026-04-14T00:00:00.000Z	2026-04-14T00:00:00.000Z	  

Delete Book

Are you sure, you want to delete book Biopic?

Cancel Delete

< 1 >

Fig 7.3.5: Deleting a Book Record

4	Pride and Prejudice	relationships among the Bennet sisters, in particular Elizabeth Bennet the middle daughter, and the wealthy Mr. Darcy. Austen satirizes the social classes of the English gentry through a witty and ironic narrative voice.	1813-01-28T00:00:00.000Z	Jane Austen	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
5	Harry Potter and the Prisoner of Azkaban	Harry's third year of studies at Hogwarts is threatened by Sirius Black's escape from Azkaban prison. Apparently, it is a dangerous wizard who was an accomplice of Lord Voldemort and who will try to take revenge on Harry Potter.	1999-07-08T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
6	Harry Potter and the Goblet of Fire	Hogwarts prepares for the Triwizard Tournament, in which three schools of wizardry will compete. To everyone's surprise, Harry Potter is chosen to participate in the competition, in which he must fight dragons, enter the water and face his greatest fears.	2000-07-08T00:00:00.000Z	J.K. Rowling (Joanne Kathleen Rowling)	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
7	The Hitchhiker's Guide to the Galaxy	A comic science fiction comedy series created by Douglas Adams. The story follows the comedic misadventures of Arthur Dent, a hapless Englishman, following the destruction of the Earth by the Vogons, a race of unpleasant bureaucratic aliens. Arthur escapes with his friend Ford Prefect, who reveals himself to be an undercover researcher for the titular Hitchhiker's Guide to the Galaxy, a galactic encyclopedia containing information about anything and everything.	1979-10-12T00:00:00.000Z	Douglas Adams	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
8	Frankenstein; or, The Modern Prometheus	A Gothic novel by Mary Shelley that tells the story of Victor Frankenstein, a young scientist who creates a grotesque creature in an unorthodox scientific experiment. Frankenstein is horrified by his creation and abandons it, but the creature seeks revenge. The novel explores themes of scientific responsibility, creation versus destruction, and the nature of good and evil.	1818-03-03T00:00:00.000Z	Mary Shelley	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
9	The Lord of the Rings: The Fellowship of the Ring	The first book in J.R.R. Tolkien's epic fantasy trilogy, The Lord of the Rings. The Fellowship of the Ring follows hobbit Frodo Baggins as he inherits the One Ring, an evil artifact of power created by the Dark Lord Sauron. Frodo embarks on a quest to destroy the Ring in the fires of Mount Doom, accompanied by a fellowship of eight companions.	1954-07-29T00:00:00.000Z	J.R.R. Tolkien	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  

Fig 7.3.6: Book Record Successfully Deleted

ID	Author	Birthday	Description	Created Date	Updated Date	Actions
1	J.K. Rowling (Joanne Kathleen Rowling)	1965-07-31T00:00:00.000Z	J.K. Rowling is a British author best known for writing the Harry Potter fantasy series. The series has won multiple awards and sold over 500 million copies. It has become the best-selling book series in history. Rowling has also written other novels, including The Casual Vacancy and The Girl on the Train.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
3	Jane Austen	1775-12-16T00:00:00.000Z	Jane Austen was an English novelist, who explored themes of the English language.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
4	Harper Lee	1960-07-11T00:00:00.000Z	Harper Lee was an American novelist best known for her Pulitzer Prize-winning novel To Kill a Mockingbird. The novel explores themes of racial injustice and the importance of compassion. Lee published a sequel, Go Set a Watchman, in 2015.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
5	J.R.R. Tolkien	1954-07-29T00:00:00.000Z	J.R.R. Tolkien was a British philologist and writer best known for his fantasy novels The Hobbit and The Lord of the Rings. Tolkien's works have had a profound influence on the fantasy genre and popular culture.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
6	Mary Shelley	1818-03-03T00:00:00.000Z	Mary Shelley was a British novelist, playwright, and short story writer, the daughter of Mary Wollstonecraft Godwin and the wife of poet Percy Bysshe Shelley. Frankenstein, or, The Modern Prometheus (1818) is her most famous work.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
7	Douglas Adams	1979-10-12T00:00:00.000Z	Douglas Adams was an English science fiction writer, satirist, humorist, dramatist, screenwriter, and occasional actor. He is best known for the Hitchhiker's Guide to the Galaxy comedy series, which inspired a radio comedy, several books, stage shows, comic books, a 1981 TV series, a 1984 video game, a 2005 feature film, and a 2008 sequel film.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	  
8	Snehal Pawar	2026-04-16T00:00:00.000Z	This is only for testing purpose	2026-04-14T00:00:00.000Z	2026-04-14T00:00:00.000Z	  

Fig 7.3.7: Deleting an Author Record

ID	Author	Birthday	Description	Created Date	Updated Date	Actions
1	J.K. Rowling (Joanne Kathleen Rowling)	1965-07-31T00:00:00.000Z	J.K. Rowling is a British author best known for writing the Harry Potter fantasy series. The series has won multiple awards and sold over 500 million copies, becoming the best-selling book series in history. Rowling has also written other novels, including The Casual Vacancy and the Cormoran Strike crime series under the pen name Robert Galbraith.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
3	Jane Austen	1775-12-16T00:00:00.000Z	Jane Austen was an English novelist known for her wit, social commentary, and romantic stories. Her six major novels, which explore themes of love, marriage, and money, have earned her a place as one of the greatest writers in the English language.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
4	Harper Lee	1960-07-11T00:00:00.000Z	Harper Lee was an American novelist best known for her Pulitzer Prize-winning novel To Kill a Mockingbird. The novel explores themes of racial injustice and the importance of compassion. Lee published a sequel, Go Set a Watchman, in 2015.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
5	J.R.R. Tolkien	1954-07-29T00:00:00.000Z	J.R.R. Tolkien was a British philologist and writer best known for his fantasy novels The Hobbit and The Lord of the Rings. Tolkien's works have had a profound influence on the fantasy genre and popular culture.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
6	Mary Shelley	1818-03-03T00:00:00.000Z	Mary Shelley was a British novelist, playwright, and short story writer, the daughter of Mary Wollstonecraft Godwin and the wife of poet Percy Bysshe Shelley. Frankenstein, or, The Modern Prometheus (1818) is her most famous work.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]
7	Douglas Adams	1979-10-12T00:00:00.000Z	Douglas Adams was an English science fiction writer, satirist, humorist, dramatist, screenwriter, and occasional actor. He is best known for the Hitchhiker's Guide to the Galaxy comedy series, which inspired a radio comedy, several books, stage shows, comic books, a 1981 TV series, a 1984 video game, a 2005 feature film, and a 2008 sequel film.	2024-05-29T00:00:00.000Z	2024-05-29T00:00:00.000Z	[Edit] [Share] [Delete]

Fig 7.3.8: Author Record Successfully Deleted

7.4 Logging and Monitoring Report

CloudWatch logs were used to monitor application activity and debug issues. Logs generated by EC2 instances were successfully captured and stored in log groups, providing visibility into system behavior.

Timestamp	Message
No older events at this moment. Retry	
2026-04-14 10:40:14 [INFO]	Connected to MySQL Database
2026-04-14 10:42:48 [INFO]	BooksController [CREATE] - title: Biopic, description: This is for Testing Purpose, releaseDate: 2026-12-01, pages: 50, authorId: 9
2026-04-14 10:42:48 [INFO]	Book created successfully, books count: 9
2026-04-14 10:42:54 [INFO]	BooksController [GET]
2026-04-14 10:42:54 [INFO]	Books count: 9
2026-04-14 10:45:06 [INFO]	AuthorsController [GET]
2026-04-14 10:45:06 [INFO]	Authors count: 7
2026-04-14 10:45:15 [INFO]	BooksController [DELETE] - bookId: 11
2026-04-14 10:45:15 [INFO]	Book deleted successfully, books count: 8
No newer events at this moment. Auto retry paused. Resume	

Fig 7.4.1: Application Logs in CloudWatch

7.5 Final Application Access Report

The application was successfully accessed using a custom domain name and HTTPS protocol. This confirms that DNS configuration, SSL certificate, and CloudFront distribution are working correctly.

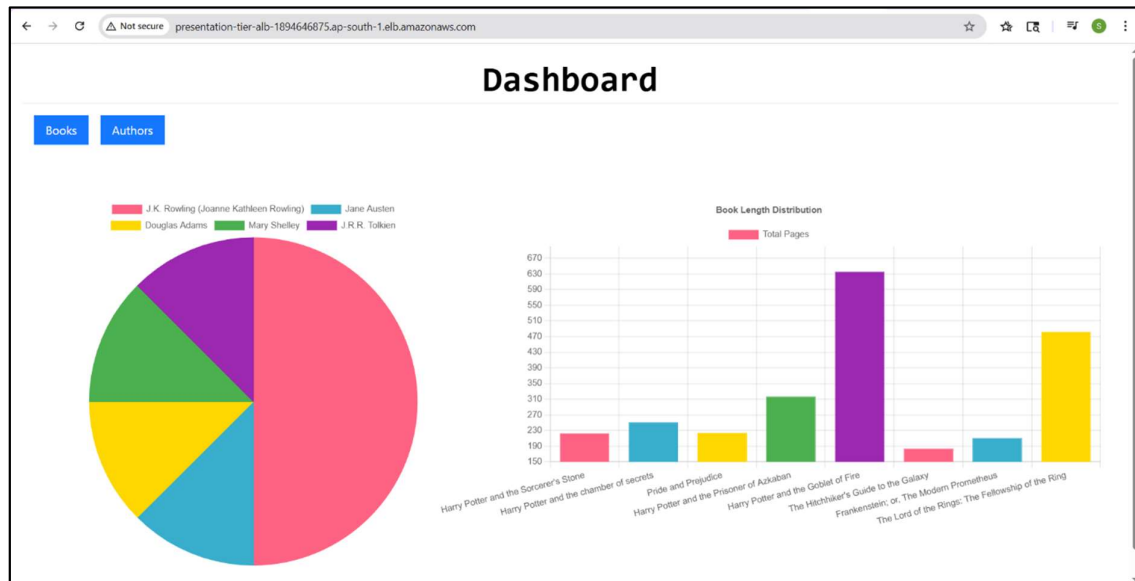


Fig 7.5.1: Application Access via Application Load Balancer DNS

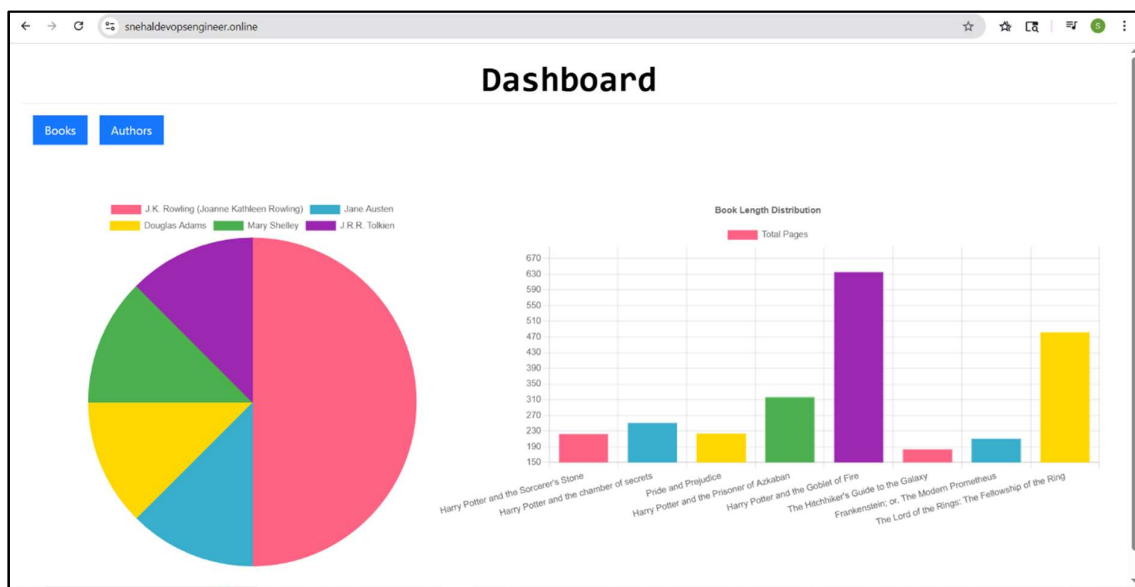


Fig 7.5.2: Application Access via Custom Domain

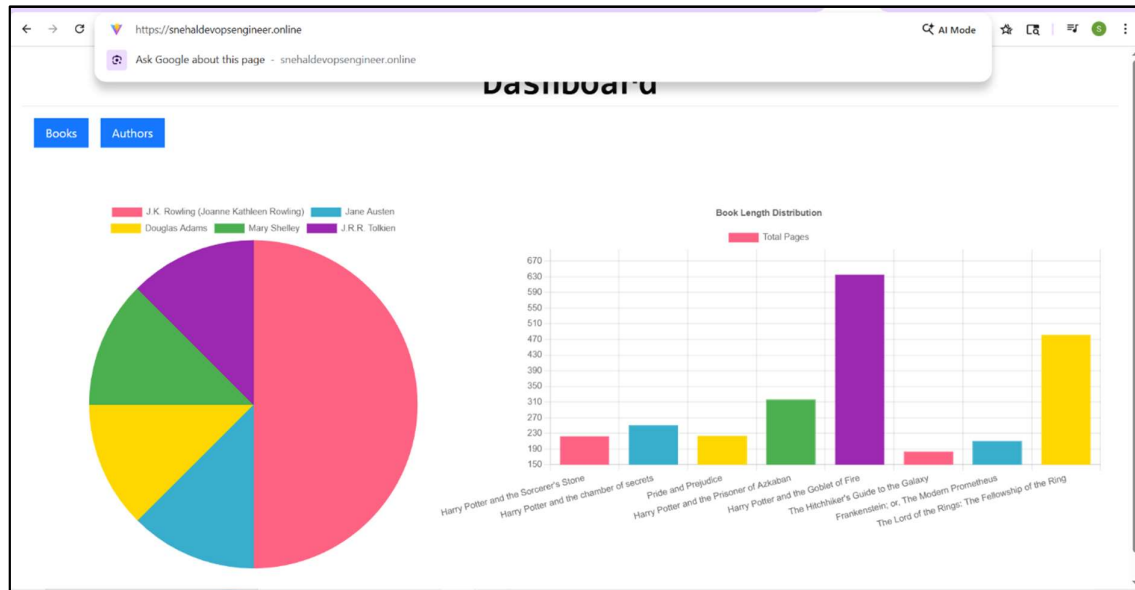


Fig 7.5.3: Secure Application Access via HTTPS

CONCLUSION

The project successfully demonstrates the design and deployment of a scalable and highly available three-tier architecture using Amazon Web Services. By separating the system into presentation, application, and data layers, the architecture ensures better performance, maintainability, and security.

The implementation of key AWS services such as EC2, Application Load Balancer, Auto Scaling Group, RDS, CloudFront, Route53, and CloudWatch enabled the development of a robust cloud-based infrastructure. The system efficiently handles user requests, distributes traffic across multiple instances, and maintains availability even in case of instance failure.

The integration of CloudWatch logging provides centralized monitoring, making it easier to track application behavior and debug issues. Additionally, the use of CloudFront improves content delivery speed and enhances user experience through reduced latency.

Overall, the project provides practical exposure to real-world cloud deployment practices and demonstrates how modern applications can be built using cloud and DevOps principles. The objectives of scalability, reliability, performance, and monitoring have been successfully achieved.

FUTURE SCOPE

Although the current system meets the project objectives, several enhancements can be implemented to further improve the architecture and align it with industry standards.

In future, a CI/CD pipeline can be integrated using tools such as Jenkins or GitHub Actions to automate the build and deployment process. This will improve development efficiency and reduce manual effort.

Containerization using Docker and orchestration using Kubernetes can be implemented to make the system more portable and scalable. This will help in managing applications more efficiently in a production environment.

Advanced security features such as Web Application Firewall (WAF), IAM role optimization, and network security improvements can be added to strengthen system protection against threats.

Caching mechanisms such as Redis can be introduced to improve application performance and reduce database load. This will enhance response time and overall efficiency.

Monitoring can be further enhanced by creating custom dashboards and alerts in CloudWatch, enabling proactive system management.

These improvements will transform the system into a more advanced, production-ready cloud solution.

BIBLIOGRAPHY AND REFERENCES

The following resources were referred to during the development and implementation of this project:

- Amazon Web Services Documentation
<https://docs.aws.amazon.com>
- AWS EC2 Documentation
<https://docs.aws.amazon.com/ec2>
- AWS RDS Documentation
<https://docs.aws.amazon.com/rds>
- AWS CloudFront Documentation
<https://docs.aws.amazon.com/cloudfront>
- AWS Route53 Documentation
<https://docs.aws.amazon.com/route53>
- AWS CloudWatch Documentation
<https://docs.aws.amazon.com/cloudwatch>
- Node.js Official Documentation
<https://nodejs.org>
- React.js Official Documentation
<https://reactjs.org>
- MySQL Documentation
<https://dev.mysql.com/doc>
- GitHub Repository (Project Source Code)
- Internship Learning Materials provided by Exlearn Technologies